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UPPER SIXTH — FURTHER MATHEMATICS

## CHAPTER 13

# LINEAR TRANSFORMATION

**SUGGESTED TIME: 6 PERIODS**

IN THIS CHAPTER

This chapter studies linear transformations of the plane and of space: their algebraic properties, their kernels and images, their inverses and invariant points, the way they act on lines and planes, and the eigenvalues and eigenvectors that reveal their hidden structure. It provides the essential bridge between matrix algebra and geometric reasoning required for the GCE Advanced Level examination.

- By the end of this chapter I can define a linear transformation and verify whether a given mapping is linear
- By the end of this chapter I can state and use the properties of linear transformations (preservation of addition, scalar multiplication, the zero vector and linear combinations)
- By the end of this chapter I can determine the kernel and the image of a linear transformation and interpret them geometrically
- By the end of this chapter I can decide whether a linear transformation is injective, surjective or bijective using the kernel and image
- By the end of this chapter I can find the inverse of an invertible linear transformation and determine its invariant points and invariant lines
- By the end of this chapter I can find the images of lines and planes under a linear transformation


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THIS SIGNED COURSE CONTAINS

**COURSE** complete, illustrated, step by step

**METHODS** and worked examples

**EXERCISES** exam style + solutions

**SUMMARY SHEET** the essentials to remember

CONTENTS

1. Linear Transformations and Their Properties
2. Kernel and Image of a Linear Transformation
3. Inverse of a Linear Transformation and Invariant Points
4. Images of Lines and Planes
5. Eigenvalues and Eigenvectors
6. Integration activity
7. Methods and worked examples
8. Exercises
9. Solutions to the exercises
10. Summary sheet

CHAPTER OBJECTIVES

- By the end of this chapter I can define a linear transformation and verify whether a given mapping is linear
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- By the end of this chapter I can find the inverse of an invertible linear transformation and determine its invariant points and invariant lines
- By the end of this chapter I can find the images of lines and planes under a linear transformation
- By the end of this chapter I can compute the eigenvalues and eigenvectors of a linear transformation or of its associated matrix
- By the end of this chapter I can use eigenvectors to describe the geometric effect of a transformation (stretching along fixed directions)
- By the end of this chapter I can solve GCE-style problems combining matrices, linear transformations and geometry

**PREREQUISITES**

- Vectors in the plane and in space: coordinates, addition, scalar multiplication, collinearity (Lower Sixth)
- Matrices: operations, determinant of a 2×2 and 3×3 matrix, inverse of a matrix (Lower Sixth / Term 1)
- Solving systems of linear equations by substitution, elimination and row reduction
- Equations of lines and planes in 2D and 3D (vector, parametric and Cartesian forms)
- Mappings and functions: domain, image set, injectivity, surjectivity, composition (Lower Sixth)

# 1 LINEAR TRANSFORMATIONS AND THEIR PROPERTIES

When a graphic designer in Douala enlarges a photo of the Reunification Monument, or when a surveyor in Bamenda projects a hillside onto a flat map, every point of the original figure is moved to a new position by a precise rule. Some rules preserve straight lines, parallelism and the origin; others do not. How can we describe such rules algebraically, predict which points remain fixed, and find the special directions that are merely stretched or compressed? These questions lead us to the powerful notion of a **linear transformation**, a tool that unifies geometry, matrices and systems of equations.

## 1.1 Definition and Fundamental Axioms

Let $E$ and $F$ be two vector spaces over the same field $\mathbb{R}$. A mapping $T : E \to F$ is called a **linear transformation** (or **linear map**) if it satisfies two axioms.

**DEFINITION** **Linear Transformation**

A map $T : E \to F$ is linear if and only if, for all vectors $\vec{u}, \vec{v} \in E$ and every scalar $k \in \mathbb{R}$:

1. Additivity: $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$;
2. Homogeneity: $T(k\vec{u}) = k\,T(\vec{u})$.

These two conditions can be combined into a single, extremely useful criterion.

**PROPERTY** **Combined Linearity Condition**

$T$ is linear $\iff$ $T(a\vec{u} + b\vec{v}) = a\,T(\vec{u}) + b\,T(\vec{v})$ for all $\vec{u}, \vec{v} \in E$ and all scalars $a, b$.

**Proof.**

If $T$ is linear, then $T(a\vec{u} + b\vec{v}) = T(a\vec{u}) + T(b\vec{v}) = aT(\vec{u}) + bT(\vec{v})$. Conversely, taking $a = b = 1$ gives additivity, and taking $b = 0$ gives homogeneity. □

This combined condition is the most efficient tool for testing whether a given mapping is linear. It also shows immediately that a linear transformation preserves all linear combinations:

$$T\Big(\sum_{i=1}^{n} \alpha_i \vec{v}_i\Big) = \sum_{i=1}^{n} \alpha_i T(\vec{v}_i).$$

## 1.2 Immediate Consequences

**PROPERTY** **Fundamental Properties of Linear Maps**

Let $T : E \to F$ be linear. Then:

1. $T(\vec{0}_E) = \vec{0}_F$ (the zero vector maps to the zero vector);
2. $T(-\vec{u}) = -T(\vec{u})$ for all $\vec{u} \in E$;
3. $T(\vec{u} - \vec{v}) = T(\vec{u}) - T(\vec{v})$;
4. $T$ preserves linear combinations: $T\left(\sum_i \alpha_i \vec{v}_i\right) = \sum_i \alpha_i T(\vec{v}_i)$.

**Proof.**
(1) $T(\vec{0}) = T(0 \cdot \vec{0}) = 0 \cdot T(\vec{0}) = \vec{0}$. (2) $T(-\vec{u}) = T((-1)\vec{u}) = (-1)T(\vec{u}) = -T(\vec{u})$. (3) $T(\vec{u} - \vec{v}) = T(\vec{u} + (-\vec{v})) = T(\vec{u}) + T(-\vec{v}) = T(\vec{u}) - T(\vec{v})$. (4) follows by induction from additivity and homogeneity. □

**WARNING** **The Zero-Vector Test — A Quick Way to Spot Non-Linearity**

If a mapping $T$ does not send the zero vector to the zero vector, it is not linear. This is the very first check you should perform in an examination.

- $T(x, y) = (x + 1, y)$ gives $T(0, 0) = (1, 0) \neq (0, 0)$, so $T$ is NOT linear.
- $T(x, y) = (x^2, y)$ gives $T(0, 0) = (0, 0)$: it passes the zero test, but fails homogeneity (see below). The zero test can only prove non-linearity, never linearity.

## 1.3 Standard Examples in the Plane

The most familiar linear transformations act on the Euclidean plane $\mathbb{R}^2$. We denote the standard basis vectors by $\vec{i} = (1, 0)$ and $\vec{j} = (0, 1)$.

**EXEMPLE** **Identity and Zero Transformations**

- Identity: $I(\vec{u}) = \vec{u}$. Matrix: $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.
- Zero map: $O(\vec{u}) = \vec{0}$. Matrix: $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.

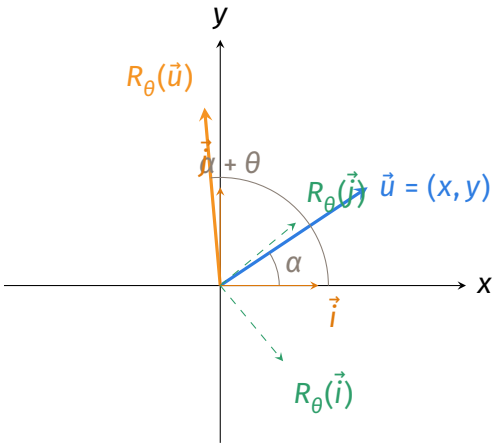
Both are trivially linear.

**EXEMPLE** **Homothety (Enlargement) of Ratio $k$**

$H_k(x, y) = (kx, ky)$ stretches every vector by the factor $k$ from the origin. Matrix: $\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} = kI$.

**EXEMPLE** **Rotation about the Origin through Angle $\theta$**

$R_\theta(x, y) = (x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta)$. Matrix: $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.




Rotation through $\theta$: the images of $\vec{i}$ and $\vec{j}$ are the columns of the rotation matrix.

**EXEMPLE** **Reflection in a Line through the Origin**

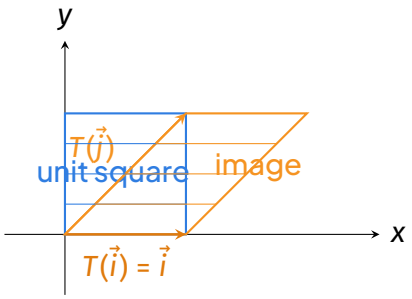
Reflection in the line $y = x$: $S(x, y) = (y, x)$, matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Reflection in the $x$-axis: $S_x(x, y) = (x, -y)$, matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Reflection in the line making angle $\alpha$ with the positive $x$-axis: matrix $\begin{pmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{pmatrix}$.

**EXEMPLE** **Orthogonal Projection onto a Line through the Origin**

Projection onto the $x$-axis: $P_x(x, y) = (x, 0)$, matrix $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. Projection onto the line $y = mx$: matrix $\frac{1}{1+m^2}\begin{pmatrix} 1 & m \\ m & m^2 \end{pmatrix}$.

**EXEMPLE** **Shear Transformations**

Horizontal shear: $S_h(x, y) = (x + ky, y)$, matrix $\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$. Vertical shear: $S_v(x, y) = (x, y + kx)$, matrix $\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$.




Horizontal shear $T(x, y) = (x + y, y)$: the unit square becomes a parallelogram; horizontal lines stay horizontal and parallel.

## 1.4 Matrix Representation of a Linear Transformation in the Plane

All the examples above have the form $T(x, y) = (ax + by,\ cx + dy)$. This is no accident.

**PROPERTY** **Matrix of a Linear Transformation of $\mathbb{R}^2$**

Every linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ can be written uniquely as

$$T(x, y) = (ax + by,\ cx + dy)$$

for some real numbers $a, b, c, d$. The **matrix of $T$** relative to the standard basis $\{\vec{i}, \vec{j}\}$ is

$$\mathbf{M}_T = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

whose first column is $T(\vec{i})$ and whose second column is $T(\vec{j})$.

**Proof.**

Write $T(\vec{i}) = (a, c)$ and $T(\vec{j}) = (b, d)$. For any $(x, y) = x\vec{i} + y\vec{j}$, linearity gives

$$T(x, y) = xT(\vec{i}) + yT(\vec{j}) = x(a, c) + y(b, d) = (ax + by,\ cx + dy).$$

Conversely, any map of this form satisfies the combined linearity condition, since each component is a linear form in $(x, y)$. □

**METHOD** **Building the Matrix from a Geometric Description**

To find the matrix of a linear transformation $T$ described geometrically:

1. Determine the images of the basis vectors $\vec{i} = (1, 0)$ and $\vec{j} = (0, 1)$.
2. Write these images as column vectors.
3. The matrix is $\mathbf{M}_T = \left[\, T(\vec{i})\ \ T(\vec{j}) \,\right]$.

**WORKED EXAMPLE 1** **Finding the Matrix of a Rotation through $60^\circ$**

Statement: Find the matrix of the rotation $R_{60^\circ}$ about the origin.

**SOLUTION**

Solution: $\cos 60^\circ = \frac{1}{2}$ and $\sin 60^\circ = \frac{\sqrt{3}}{2}$. $R_{60^\circ}(\vec{i}) = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ and $R_{60^\circ}(\vec{j}) = \left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$. Hence

$$\mathbf{M} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}.$$

## 1.5 Extension to Three-Dimensional Space

Everything generalises naturally to $\mathbb{R}^3$ with $\vec{i} = (1, 0, 0)$, $\vec{j} = (0, 1, 0)$, $\vec{k} = (0, 0, 1)$.

**PROPERTY** **Matrix of a Linear Transformation of $\mathbb{R}^3$**

Every linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ has the form

$$T(x, y, z) = (a_{11}x + a_{12}y + a_{13}z,\ a_{21}x + a_{22}y + a_{23}z,\ a_{31}x + a_{32}y + a_{33}z)$$

and is represented by the $3 \times 3$ matrix

$$\mathbf{M}_T = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \left[\, T(\vec{i})\ \ T(\vec{j})\ \ T(\vec{k}) \,\right].$$

Common examples in $\mathbb{R}^3$:

- Rotation about the $z$-axis through $\theta$: $\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$.
- Reflection in the plane $z = 0$: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$.
- Homothety of ratio $k$: $k\mathbf{I}_3$.

## 1.6 Operations on Linear Transformations

Linear transformations can be added, multiplied by scalars, and composed; these operations correspond exactly to the usual matrix operations.

**PROPERTY** **Algebra of Linear Transformations**

Let $T, S : E \to F$ be linear and $\lambda \in \mathbb{R}$.

1. Sum: $(T+S)(\vec{u}) = T(\vec{u}) + S(\vec{u})$ is linear, and $\mathbf{M}_{T+S} = \mathbf{M}_T + \mathbf{M}_S$.
2. Scalar multiple: $(\lambda T)(\vec{u}) = \lambda\, T(\vec{u})$ is linear, and $\mathbf{M}_{\lambda T} = \lambda \mathbf{M}_T$.
3. Composition: if $T : E \to F$ and $S : F \to G$ are linear, then $S \circ T : E \to G$, defined by $(S \circ T)(\vec{u}) = S(T(\vec{u}))$, is linear, and $\mathbf{M}_{S\circ T} = \mathbf{M}_S \mathbf{M}_T$.

**WARNING** **Order of Composition**

$S \circ T$ means apply $T$ first, then $S$. The matrix is $\mathbf{M}_S\mathbf{M}_T$. In general $\mathbf{M}_S\mathbf{M}_T \neq \mathbf{M}_T\mathbf{M}_S$: composition of transformations, like matrix multiplication, is not commutative.

**EXEMPLE** **Composition of Two Shears**

Let $T_1(x, y) = (x + y, y)$ with matrix $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $T_2(x, y) = (x, y + x)$ with matrix $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$.

$$\mathbf{M}_{T_2\circ T_1} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \mathbf{M}_{T_1\circ T_2} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

The two composites are different.

## 1.7 Recognising Non-Linear Maps — GCE Examination Traps

A mapping $T : \mathbb{R}^2 \to \mathbb{R}^2$ given by algebraic formulas is **linear** if and only if each component is a **homogeneous linear polynomial** in the input variables: a sum of terms of the form $ax_i$, with no constant term, no products and no powers of variables.

**METHOD** **How to Test Linearity Quickly**

1. Zero test: compute $T(\vec{0})$. If $T(\vec{0}) \neq \vec{0}$, stop: $T$ is not linear.
2. Form inspection: constant terms, products ($xy$, $yz$), powers ($x^2$), or non-linear functions ($\sin x$, $e^x$, $\sqrt{x}$, $|x|$) $\Rightarrow$ not linear.
3. Otherwise, verify the combined condition $T(a\vec{u} + b\vec{v}) = aT(\vec{u}) + bT(\vec{v})$ algebraically, or exhibit the matrix.

**WORKED EXAMPLE 2** **Identifying Linear and Non-Linear Maps**

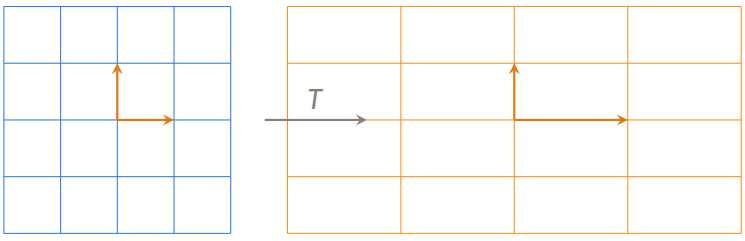
Statement: Which of the following transformations of $\mathbb{R}^2$ are linear?

1. $T_1(x, y) = (2x - y,\ 3x + 4y)$
2. $T_2(x, y) = (x + 1,\ y - 2)$
3. $T_3(x, y) = (x^2,\ y)$
4. $T_4(x, y) = (x + y,\ xy)$
5. $T_5(x, y) = (0,\ 0)$

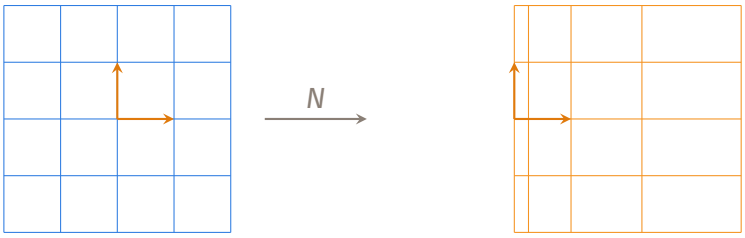
**SOLUTION**

Solution:

1. Linear. Each component is a homogeneous linear polynomial; matrix $\begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}$.
2. Not linear. $T_2(0, 0) = (1, -2) \neq (0, 0)$ (constant terms).
3. Not linear. $T_3(0, 0) = (0, 0)$, but $T_3(2, 0) = (4, 0)$ while $2T_3(1, 0) = (2, 0)$: homogeneity fails (term $x^2$).
4. Not linear. $T_4(1, 1) = (2, 1)$ while $T_4(1, 0) + T_4(0, 1) = (1, 0) + (1, 0) = (2, 0)$: additivity fails (term $xy$).
5. Linear. This is the zero transformation.




original grid  image under $T(x, y) = (2x, y)$




original grid  image under $N(x, y) = (x^2, y)$

Top: a linear map sends a rectangular grid to a parallelogram grid. Bottom: the non-linear map $N(x, y) = (x^2, y)$ collapses the lines $x = c$ and $x = -c$ onto the same line and distorts the spacing.

**KEY POINTS** **Key Points to Remember**

- $T$ is linear $\iff T(a\vec{u} + b\vec{v}) = aT(\vec{u}) + bT(\vec{v})$ for all vectors and scalars.
- Every linear map sends $\vec{0}$ to $\vec{0}$; if $T(\vec{0}) \neq \vec{0}$, $T$ is not linear.
- In $\mathbb{R}^2$, $T(x, y) = (ax + by,\ cx + dy)$ with matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ whose columns are $T(\vec{i})$ and $T(\vec{j})$.
- In $\mathbb{R}^3$, the matrix is $3 \times 3$ with columns $T(\vec{i}), T(\vec{j}), T(\vec{k})$.
- Composition corresponds to matrix multiplication: $\mathbf{M}_{S \cdot T} = \mathbf{M}_S \mathbf{M}_T$.
- Examination trap: constant terms, squares, products $(xy)$ or transcendental functions in the formula $\Rightarrow$ NOT linear.

**EXERCISE 1** **Practice Exercise 1**

1. Determine whether the following maps are linear. Justify your answer.

   (a) $T(x, y) = (3x - 2y,\ x + 5y)$

(b) $T(x, y) = (x + 2,\ y - 3)$

(c) $T(x, y) = (x^2 + y^2,\ 0)$

(d) $T(x, y) = (|x|,\ y)$

2. Find the matrix of the reflection of $\mathbb{R}^2$ in the line $y = -x$.

3. A linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ satisfies $T(1, 2) = (3, 4)$ and $T(2, 1) = (5, 6)$. Find $T(x, y)$ and its matrix.

4. Let $R_\theta$ be the rotation through $\theta$ and $S$ the reflection in the $x$-axis. Write the matrices of $R_\theta \circ S$ and $S \circ R_\theta$. Are they equal?

**SOLUTION** **Exercise 1**

1. (a) Linear: homogeneous linear components; matrix $\begin{pmatrix} 3 & -2 \\ 1 & 5 \end{pmatrix}$. (b) Not linear: $T(0, 0) = (2, -3) \neq (0, 0)$. (c) Not linear: $T(2, 0) = (4, 0)$ but $2T(1, 0) = (2, 0)$; homogeneity fails. (d) Not linear: $T(-1, 0) = (1, 0)$ but $-T(1, 0) = (-1, 0)$; the absolute value breaks homogeneity.

2. $(1, 0) \mapsto (0, -1)$ and $(0, 1) \mapsto (-1, 0)$, so the matrix is $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$.

3. Write $(x, y) = \alpha(1, 2) + \beta(2, 1)$. Solving $\alpha + 2\beta = x$, $2\alpha + \beta = y$ gives $\alpha = \dfrac{2y - x}{3}$, $\beta = \dfrac{2x - y}{3}$. Then

$$\begin{aligned} T(x, y) &= \alpha(3, 4) + \beta(5, 6) \\ &= \tfrac{1}{3}\big(3(2y - x) + 5(2x - y),\ 4(2y - x) + 6(2x - y)\big) \\ &= \tfrac{1}{3}(7x + y,\ 8x + 2y). \end{aligned}$$

Matrix: $\dfrac{1}{3}\begin{pmatrix} 7 & 1 \\ 8 & 2 \end{pmatrix}$. Check: $T(1, 2) = \tfrac{1}{3}(9, 12) = (3, 4)$ and $T(2, 1) = \tfrac{1}{3}(15, 18) = (5, 6)$. □

4. $\mathbf{M}_{R_\theta \circ S} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$; $\mathbf{M}_{S \circ R_\theta} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ -\sin\theta & -\cos\theta \end{pmatrix}$. They are not equal unless $\sin\theta = 0$.

# 2 KERNEL AND IMAGE OF A LINEAR TRANSFORMATION

A projection of $\mathbb{R}^3$ onto a plane squashes a whole direction onto the origin: infinitely many vectors share the same image. To understand a linear transformation we must answer two questions: which vectors are sent to zero? and which vectors can be reached as images? The answers are the **kernel** and the **image**.

## 2.1 Definitions

**DEFINITION** **Kernel and Image**

Let $T : E \to F$ be a linear transformation.

1. The **kernel** of $T$ is the set of all vectors of $E$ mapped to $\vec{0}_F$:

$$\ker T = \{\vec{u} \in E : T(\vec{u}) = \vec{0}_F\}.$$

2. The **image** of $T$ is the set of all vectors of $F$ that are images of vectors of $E$:

$$\operatorname{Im} T = \{T(\vec{u}) : \vec{u} \in E\}.$$

**PROPERTY** **Kernel and Image are Subspaces**

$\ker T$ is a vector subspace of $E$ and $\operatorname{Im} T$ is a vector subspace of $F$. In particular both contain the zero vector.

**Proof.**

$T(\vec{0}_E) = \vec{0}_F$, so $\vec{0}_E \in \ker T$. If $\vec{u}, \vec{v} \in \ker T$ and $a, b \in \mathbb{R}$, then $T(a\vec{u} + b\vec{v}) = aT(\vec{u}) + bT(\vec{v}) = \vec{0}$, so $a\vec{u} + b\vec{v} \in \ker T$. Similarly, if $\vec{w}_1 = T(\vec{u}_1)$ and $\vec{w}_2 = T(\vec{u}_2)$, then $a\vec{w}_1 + b\vec{w}_2 = T(a\vec{u}_1 + b\vec{u}_2) \in \operatorname{Im} T$. □

**PROPERTY** **The Image is Spanned by the Columns**

If $T : \mathbb{R}^2 \to \mathbb{R}^2$ has matrix $\mathbf{M}_T = \left[\, T(\vec{i}) \;\; T(\vec{j}) \,\right]$, then

$$\operatorname{Im} T = \operatorname{span}\{T(\vec{i}),\, T(\vec{j})\},$$

the set of all linear combinations of the columns of $\mathbf{M}_T$. The same holds in $\mathbb{R}^3$ with the three columns.

**Proof.**

$T(x, y) = xT(\vec{i}) + yT(\vec{j})$, so every image is a linear combination of the two columns, and conversely every such combination is an image. □

**METHOD** **Finding Kernel and Image from the Matrix**

Given $T(x, y) = (ax + by,\ cx + dy)$:

1. Kernel: solve the homogeneous system $ax + by = 0$, $cx + dy = 0$. The solution set is $\ker T$.
2. Image: take the span of the columns $T(\vec{i}) = (a, c)$ and $T(\vec{j}) = (b, d)$; reduce to a simplest description (a point, a line through $O$, or the whole plane).

**WORKED EXAMPLE 3** **Kernel and Image of a Projection**

Statement: Let $T(x, y) = (x + y,\ 2x + 2y)$. Determine $\ker T$ and $\operatorname{Im} T$.

**SOLUTION**

Solution: Kernel: solve $x + y = 0$ and $2x + 2y = 0$; both reduce to $y = -x$. Hence

$$\ker T = \{(t, -t) : t \in \mathbb{R}\} = \operatorname{span}\{(1, -1)\},$$

the line $y = -x$. Image: the columns are $T(\vec{i}) = (1, 2)$ and $T(\vec{j}) = (1, 2)$, so

$$\operatorname{Im} T = \operatorname{span}\{(1, 2)\} = \{(s, 2s) : s \in \mathbb{R}\},$$

the line $y = 2x$. Note that $\dim \ker T + \dim \operatorname{Im} T = 1 + 1 = 2 = \dim \mathbb{R}^2$.

## 2.2 The Rank–Nullity Theorem

**DEFINITION** **Rank and Nullity**

The **rank** of $T$ is $\operatorname{rank}(T) = \dim(\operatorname{Im} T)$; the **nullity** of $T$ is $\dim(\ker T)$.

**PROPERTY** **Rank–Nullity Theorem**

For any linear transformation $T : E \to F$ with $E$ finite-dimensional,

$$\dim(\ker T) + \dim(\operatorname{Im} T) = \dim E.$$

In the plane: $\dim \ker T + \operatorname{rank}(T) = 2$; in space: $\dim \ker T + \operatorname{rank}(T) = 3$.

**EXEMPLE** **The Three Possibilities in the Plane**

For $T : \mathbb{R}^2 \to \mathbb{R}^2$:

- rank$(T) = 2$: $\ker T = \{\vec{0}\}$, $\operatorname{Im} T = \mathbb{R}^2$ ($T$ is invertible);
- rank$(T) = 1$: $\ker T$ and $\operatorname{Im} T$ are both lines through $O$;
- rank$(T) = 0$: $T$ is the zero map, $\ker T = \mathbb{R}^2$.

**WORKED EXAMPLE 4** **Kernel and Image in $\mathbb{R}^3$**

Statement: Let $T(x, y, z) = (x + y + z,\ x - y,\ 2x + z)$. Find $\ker T$ and $\operatorname{Im} T$, and verify the rank–nullity theorem.

**SOLUTION**

Solution: Kernel: solve

$$\begin{cases} x + y + z = 0 \\ x - y = 0 \\ 2x + z = 0 \end{cases} \Rightarrow y = x, \quad z = -2x, \quad x + x - 2x = 0.$$

All three equations are satisfied by $(t, t, -2t)$, so $\ker T = \operatorname{span}\{(1, 1, -2)\}$, a line through $O$; nullity $= 1$. Image: the columns are $(1, 1, 2)$, $(1, -1, 0)$, $(1, 0, 1)$. Since $(1, 0, 1) = \frac{1}{2}(1, 1, 2) + \frac{1}{2}(1, -1, 0)$, the image is spanned by the first two columns, which are not proportional:

$$\operatorname{Im} T = \operatorname{span}\{(1, 1, 2),\ (1, -1, 0)\},$$

a plane through $O$; rank $= 2$. Check: $1 + 2 = 3 = \dim \mathbb{R}^3$. □

**KEY POINTS** **Key Points**

- $\ker T = \{\vec{u} : T(\vec{u}) = \vec{0}\}$: solve the homogeneous system given by the matrix.
- $\operatorname{Im} T$ is the span of the columns of $\mathbf{M}_T$.
- Rank–nullity: $\dim \ker T + \dim \operatorname{Im} T = \dim E$.
- $T$ is one-to-one (injective) $\iff \ker T = \{\vec{0}\}$.

# 3 INVERSE OF A LINEAR TRANSFORMATION AND INVARIANT POINTS

## 3.1 Invertible Linear Transformations

**DEFINITION** **Inverse Transformation**

A linear transformation $T : E \to E$ is **invertible** if there exists a map $T^{-1} : E \to E$ such that

$$T^{-1} \circ T = T \circ T^{-1} = I.$$

$T^{-1}$ is then automatically linear, and $\mathbf{M}_{T^{-1}} = \mathbf{M}_T^{-1}$.

**PROPERTY** **Invertibility Criterion**

For $T : \mathbb{R}^2 \to \mathbb{R}^2$ with matrix $\mathbf{M}$, the following are equivalent:

1. $T$ is invertible;
2. $\det \mathbf{M} \neq 0$;
3. $\ker T = \{\vec{0}\}$;
4. $\operatorname{Im} T = \mathbb{R}^2$.

When $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $ad - bc \neq 0$,

$$\mathbf{M}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

**EXEMPLE** **Inverses of the Standard Transformations**

- Rotation: $R_\theta^{-1} = R_{-\theta}$ (rotate back).
- Homothety: $H_k^{-1} = H_{1/k}$ for $k \neq 0$.
- Reflection: $S^{-1} = S$ (a reflection is its own inverse: $S \circ S = I$).
- A projection $P$ (other than $I$) is never invertible: $\det \mathbf{M}_P = 0$ and $\ker P \neq \{\vec{0}\}$.

**WORKED EXAMPLE 5** **Computing an Inverse Transformation**

Statement: Let $T(x, y) = (2x + y,\ 3x + 2y)$. Show that $T$ is invertible and find $T^{-1}(x, y)$.

**SOLUTION**

Solution: $\mathbf{M} = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$, $\det \mathbf{M} = 4 - 3 = 1 \neq 0$, so $T$ is invertible.

$$\mathbf{M}^{-1} = \frac{1}{1}\begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}, \qquad \text{hence} \qquad T^{-1}(x, y) = (2x - y,\ -3x + 2y).$$

Check: $T^{-1}(T(1, 1)) = T^{-1}(3, 5) = (6 - 5,\ -9 + 10) = (1, 1)$. □

## 3.2 Invariant (Fixed) Points

**DEFINITION** **Invariant Point**

A point $P$ with position vector $\vec{u}$ is an **invariant point** (or fixed point) of $T$ if $T(\vec{u}) = \vec{u}$: the point is left unmoved by the transformation.

Since $T(\vec{0}) = \vec{0}$, the origin is always an invariant point of a linear transformation. The interesting question is whether there are others. The invariant points are exactly the solutions of

$$(\mathbf{M} - \mathbf{I})\vec{u} = \vec{0}, \qquad \text{i.e.} \qquad \ker(T - I).$$

The set of invariant points is therefore always a subspace: $\{\vec{0}\}$, a line through $O$, a plane through $O$, or the whole space.

**METHOD** **Finding Invariant Points**

1. Write the condition $T(x, y) = (x, y)$.
2. Bring everything to one side: this gives a homogeneous system $(\mathbf{M} - \mathbf{I})\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.
3. Solve; describe the solution set geometrically.

**WORKED EXAMPLE 6** **Invariant Points of a Reflection and of a Shear**

Statement: Find the invariant points of (a) the reflection $S$ in the line $y = x$; (b) the shear

$T(x, y) = (x + 2y,\ y)$.

**SOLUTION**

Solution: (a) $S(x, y) = (y, x) = (x, y) \iff y = x$. The invariant points form the line $y = x$ — exactly the mirror line, as expected. (b) $T(x, y) = (x, y) \iff x + 2y = x \iff y = 0$. The invariant points form the $x$-axis: every point of the $x$-axis is fixed, every other point slides horizontally. □

**WARNING** **Invariant Points vs Invariant Lines**

Do not confuse the two notions:

- An invariant point satisfies $T(\vec{u}) = \vec{u}$: the point does not move.
- A line $\ell$ is an invariant line if $T(\ell) \subseteq \ell$: the line is mapped into itself, but its individual points may move along it. For the homothety $H_2$, every line through $O$ is invariant, yet the only invariant point is $O$.

**WORKED EXAMPLE 7** **A Line of Invariant Points**

Statement: Find the invariant points of $T(x, y) = (3x - 2y,\ 2x - y)$.

**SOLUTION**

Solution: $T(x, y) = (x, y)$ gives

$$\begin{cases} 3x - 2y = x \\ 2x - y = y \end{cases} \iff \begin{cases} 2x - 2y = 0 \\ 2x - 2y = 0 \end{cases} \iff x = y.$$

The invariant points form the line $y = x$. (Indeed $\mathbf{M} - \mathbf{I} = \begin{pmatrix} 2 & -2 \\ 2 & -2 \end{pmatrix}$ has rank 1, so its kernel is a line.) □

## 4 IMAGES OF LINES AND PLANES

A photograph enlarged by a printer keeps straight lines straight: this is a general fact about linear maps. We now make it precise and learn how to compute the image of a given line or plane.

### 4.1 Image of a Line

**PROPERTY** **Linear Maps Preserve Lines**

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be linear and let $\ell$ be the line through the point $A$ with direction vector $\vec{d}$, i.e. $\ell = \{A + t\vec{d} : t \in \mathbb{R}\}$. Then

$$T(\ell) = \{T(A) + t\,T(\vec{d}) : t \in \mathbb{R}\}.$$

- If $T(\vec{d}) \neq \vec{0}$, then $T(\ell)$ is the line through $T(A)$ with direction $T(\vec{d})$.
- If $T(\vec{d}) = \vec{0}$, then $T(\ell)$ collapses to the single point $T(A)$.

In particular, the image of a line through the origin is a line through the origin (or just $\{O\}$).

**Proof.**

$T(A + t\vec{d}) = T(A) + tT(\vec{d})$ by linearity; as $t$ ranges over $\mathbb{R}$ this describes exactly the stated set. □

**METHOD** **Finding the Image of a Line**

Given a line $\ell$ and a linear map $T$:

1. Write $\ell$ in parametric form: pick a point $A$ on $\ell$ and a direction vector $\vec{d}$.
2. Compute $T(A)$ and $T(\vec{d})$.
3. If $T(\vec{d}) \neq \vec{0}$, write the Cartesian equation of the line through $T(A)$ in direction $T(\vec{d})$.

Alternative for a line $ax + by = c$: express $x, y$ from the parametric form, substitute into $T$, then eliminate the parameter.

**WORKED EXAMPLE 8** **Image of a Line under a Linear Map**

Statement: Let $T(x, y) = (x + 2y,\ 3x - y)$. Find the image of the line $\ell : x + y = 1$.

**SOLUTION**

Solution: Parametrise $\ell$: $A = (1, 0)$ lies on $\ell$ and $\vec{d} = (1, -1)$ is a direction vector, so points of $\ell$ are $(1 + t,\ -t)$.

$$T(1 + t,\ -t) = \big((1 + t) + 2(-t),\ 3(1 + t) - (-t)\big) = (1 - t,\ 3 + 4t).$$

So the image is the line through $T(A) = (1, 3)$ with direction $T(\vec{d}) = (-1, 4)$. Eliminating $t$: from $x = 1 - t$ we get $t = 1 - x$, and substituting in $y = 3 + 4t$ gives

$$y = 3 + 4(1 - x) \iff 4x + y = 7.$$

Hence $T(\ell)$ is the line $4x + y = 7$. □

**WORKED EXAMPLE 9** **A Line that Collapses to a Point**

Statement: Let $T(x, y) = (x + y,\ 2x + 2y)$ (as in a previous example, $\ker T$ is the line $y = -x$). Find the image of the line $\ell : y = -x + 3$.

**SOLUTION**

Solution: Points of $\ell$: $(t,\ 3 - t)$. Then

$$T(t,\ 3 - t) = \big(t + 3 - t,\ 2t + 2(3 - t)\big) = (3, 6).$$

Every point of $\ell$ has the same image: $T(\ell) = \{(3, 6)\}$, a single point. This happens because the direction $(1, -1)$ of $\ell$ lies in $\ker T$. □

## 4.2 Image of a Plane in $\mathbb{R}^3$

**PROPERTY** **Image of a Plane**

Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be linear and let $\Pi$ be the plane through $A$ spanned by the independent directions $\vec{d}_1, \vec{d}_2$:

$$\Pi = \{A + s\vec{d}_1 + t\vec{d}_2 : s, t \in \mathbb{R}\}.$$

Then $T(\Pi) = \{T(A) + s\,T(\vec{d}_1) + t\,T(\vec{d}_2) : s, t \in \mathbb{R}\}$, which is:

- a plane if $T(\vec{d}_1)$ and $T(\vec{d}_2)$ are not proportional;
- a line if they are proportional (and not both zero);
- a point if both are zero.

**WORKED EXAMPLE 10** **Image of a Plane under a Projection**

Statement: Let $T(x, y, z) = (x,\ y,\ 0)$ be the orthogonal projection onto the plane $z = 0$. Find the image of the plane $\Pi : x + y + z = 3$.

**SOLUTION**

Solution: Points of $\Pi$: $(x, y, 3 - x - y)$. Then

$$T(x,\ y,\ 3 - x - y) = (x,\ y,\ 0).$$

As $x$ and $y$ range freely over $\mathbb{R}$, the image is the whole plane $z = 0$: $T(\Pi) = \{(x, y, 0) : x, y \in \mathbb{R}\}$. The projection flattens the tilted plane $\Pi$ onto the coordinate plane $z = 0$. □

**WARNING** **Common Mistakes**

- The image of a line not through the origin need not pass through the origin: compute $T(A)$ carefully.
- Do not forget the degenerate case: if the direction vector lies in $\ker T$, the image is a point, not a line.
- Parallel lines remain parallel (or collapse): $T$ sends direction $\vec{d}$ to $T(\vec{d})$ for every line with that direction.

# 5 EIGENVALUES AND EIGENVECTORS

Under a reflection in the line $y = x$, the vector $(1, 1)$ is fixed and the vector $(1, -1)$ is flipped to its opposite: both keep their line of action. Under a general linear map, most directions are rotated, but some special directions are preserved — the map merely stretches, compresses or reverses vectors along them. These are the **eigendirections**, and they reveal the deep structure of the transformation.

## 5.1 Definitions

**DEFINITION** **Eigenvalue and Eigenvector**

Let $T : E \to E$ be a linear transformation. A scalar $\lambda$ is an **eigenvalue** of $T$ if there exists a non-zero vector $\vec{u}$ such that

$$T(\vec{u}) = \lambda\vec{u}.$$

Such a vector $\vec{u}$ is called an **eigenvector** of $T$ associated with $\lambda$. The set

$$E_\lambda = \{\vec{u} : T(\vec{u}) = \lambda\vec{u}\} = \ker(T - \lambda I)$$

is the **eigenspace** associated with $\lambda$; it is a subspace of $E$.

**WARNING** **The Zero Vector is Never an Eigenvector**

$T(\vec{0}) = \lambda\vec{0}$ holds for every $\lambda$, so $\vec{0}$ is excluded from the definition of an eigenvector. However, $\lambda = 0$ can be an eigenvalue: it is one exactly when $\ker T \neq \{\vec{0}\}$, i.e. when $T$ is not invertible.

## 5.2 The Characteristic Equation

The condition $T(\vec{u}) = \lambda\vec{u}$ with $\vec{u} \neq \vec{0}$ means $(\mathbf{M} - \lambda\mathbf{I})\vec{u} = \vec{0}$ has a non-trivial solution, which happens exactly when the matrix $\mathbf{M} - \lambda\mathbf{I}$ is singular.

**PROPERTY** **Characteristic Equation**

The eigenvalues of $T$ with matrix $\mathbf{M}$ are the roots of the **characteristic equation**

$$\det(\mathbf{M} - \lambda\mathbf{I}) = 0.$$

For $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ this is the quadratic

$$\lambda^2 - (a + d)\lambda + (ad - bc) = 0, \qquad \text{i.e.} \qquad \lambda^2 - (\operatorname{tr}\mathbf{M})\lambda + \det\mathbf{M} = 0,$$

where $\operatorname{tr}\mathbf{M} = a + d$ is the **trace** of $\mathbf{M}$.

**Proof.**

$$\det(\mathbf{M} - \lambda\mathbf{I}) = \begin{vmatrix} a - \lambda & b \\ c & d - \lambda \end{vmatrix} = (a - \lambda)(d - \lambda) - bc = \lambda^2 - (a + d)\lambda + (ad - bc).$$ □

**METHOD** **Finding Eigenvalues and Eigenvectors of a $2 \times 2$ Matrix**

1. Compute $\operatorname{tr}\mathbf{M}$ and $\det\mathbf{M}$; write the characteristic equation $\lambda^2 - (\operatorname{tr}\mathbf{M})\lambda + \det\mathbf{M} = 0$.
2. Solve the quadratic: the roots $\lambda_1, \lambda_2$ are the eigenvalues.
3. For each eigenvalue $\lambda$, solve $(\mathbf{M} - \lambda\mathbf{I})\vec{u} = \vec{0}$ to find the eigenspace $E_\lambda$ (a line through $O$ when $\lambda$ is a simple root).
4. Check: $\lambda_1 + \lambda_2 = \operatorname{tr}\mathbf{M}$ and $\lambda_1\lambda_2 = \det\mathbf{M}$.

**WORKED EXAMPLE 11** **Eigenvalues and Eigenvectors — Two Distinct Real Eigenvalues**

Statement: Find the eigenvalues and eigenvectors of $T(x, y) = (4x + 2y,\ x + 3y)$.

**SOLUTION**

Solution: $\mathbf{M} = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$; $\operatorname{tr}\mathbf{M} = 7$, $\det\mathbf{M} = 12 - 2 = 10$. Characteristic equation: $\lambda^2 - 7\lambda + 10 =$

$0 \iff (\lambda - 2)(\lambda - 5) = 0$, so $\lambda_1 = 2, \lambda_2 = 5$. For $\lambda_1 = 2$: $(\mathbf{M} - 2\mathbf{I})\vec{u} = \vec{0}$:

$$\begin{cases} 2x + 2y = 0 \\ x + y = 0 \end{cases} \iff y = -x, \qquad E_2 = \text{span}\{(1, -1)\}.$$

For $\lambda_2 = 5$: $(\mathbf{M} - 5\mathbf{I})\vec{u} = \vec{0}$:

$$\begin{cases} -x + 2y = 0 \\ x - 2y = 0 \end{cases} \iff x = 2y, \qquad E_5 = \text{span}\{(2, 1)\}.$$

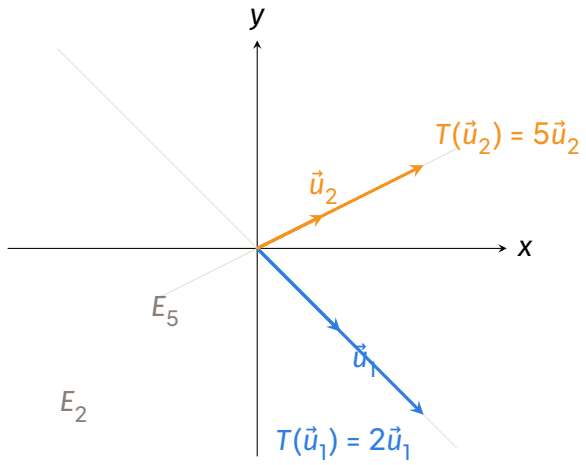
Check: $T(1, -1) = (2, -2) = 2(1, -1)$ and $T(2, 1) = (10, 5) = 5(2, 1)$. □

**WORKED EXAMPLE 12** **A Repeated Eigenvalue and No Real Eigenvalue**

Statement: Find the eigenvalues of (a) $T(x, y) = (x + 2y,\ y)$ (shear); (b) $R_{90^\circ}(x, y) = (-y,\ x)$.

**SOLUTION**

Solution: (a) $\mathbf{M} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$; characteristic equation $(1-\lambda)^2 = 0$, so $\lambda = 1$ is a double eigenvalue. Eigenspace: $2y = 0$, so $E_1 = \text{span}\{(1, 0)\}$, the $x$-axis only. A shear has a single eigendirection. (b) $\mathbf{M} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$; characteristic equation $\lambda^2 + 1 = 0$, which has no real root. A rotation through $90^\circ$ has no real eigenvalue — geometrically obvious, since no direction of the plane is preserved. □




The transformation $T(x, y) = (4x + 2y,\ \ x + 3y)$ doubles vectors along the line $E_2$ $(y = -x)$ and multiplies by 5 vectors along the line $E_5$ $(x = 2y)$.

## 5.3 Eigenvalues of the Standard Transformations

| Transformation | Eigenvalues | Eigenspaces |
|---|---|---|
| Homothety $H_k$ | $k$ (double) | the whole plane $\mathbb{R}^2$ |
| Rotation $R_\theta$ $(\theta \neq 0, \pi)$ | no real eigenvalue | — |
| Rotation $R_\pi$ (half-turn) | -1 (double) | the whole plane $\mathbb{R}^2$ |
| Reflection in a line $\ell$ | 1 and -1 | $E_1 = \ell$; $E_{-1} = \ell^\perp$ |
| Projection onto a line $\ell$ | 1 and 0 | $E_1 = \ell$; $E_0 = \ker P = \ell^\perp$ |
| Shear $T(x, y) = (x + ky,\ y)$ | 1 (double) | $E_1$ = the $x$-axis |

**PROPERTY** **Eigenvalues and Invertibility**

Let $T$ have matrix $\mathbf{M}$.

1. 0 is an eigenvalue of $T \iff \det \mathbf{M} = 0 \iff T$ is not invertible.
2. If $T$ is invertible and $\lambda$ is an eigenvalue of $T$, then $\lambda \neq 0$ and $\frac{1}{\lambda}$ is an eigenvalue of $T^{-1}$, with the same eigenvectors.
3. The eigenvalue $\lambda = 1$ occurs exactly when $T$ has invariant points other than $O$; the eigenspace $E_1$ is precisely the set of invariant points.

**Proof of (2).**

If $T(\vec{u}) = \lambda\vec{u}$ with $\vec{u} \neq \vec{0}$ and $T$ invertible, then $\lambda \neq 0$ (otherwise $\vec{u} \in \ker T$) and applying $T^{-1}$: $\vec{u} = \lambda\, T^{-1}(\vec{u})$, so $T^{-1}(\vec{u}) = \frac{1}{\lambda}\vec{u}$. □

DID YOU KNOW? **Why "Eigen"?**

The prefix eigen- is German for "own" or "characteristic": eigenvectors are the transformation's own directions. Eigenvalues are everywhere in modern science and technology: the stability of bridges and electricity pylons, the vibration modes of musical instruments, quantum mechanics (energy levels are eigenvalues), and the PageRank algorithm that orders web search results all rest on eigenvalue computations.

KEY POINTS **Key Points**

- $\lambda$ is an eigenvalue $\iff T(\vec{u}) = \lambda\vec{u}$ for some $\vec{u} \neq \vec{0} \iff \det(\mathbf{M} - \lambda\mathbf{I}) = 0$.
- For a $2 \times 2$ matrix: $\lambda^2 - (\operatorname{tr}\mathbf{M})\lambda + \det\mathbf{M} = 0$; check with $\lambda_1 + \lambda_2 = \operatorname{tr}\mathbf{M}$, $\lambda_1\lambda_2 = \det\mathbf{M}$.
- The eigenspace $E_\lambda = \ker(T - \lambda I)$ always contains $\vec{0}$; eigenvectors are its non-zero elements.
- $E_1$ is the set of invariant points; $E_0 = \ker T$.
- A rotation (other than 0 or $\pi$) has no real eigenvalue; a reflection always has 1 and -1.

EXERCISE 2 **Practice Exercise 2**

1. Let $T(x, y) = (2x + y,\ x + 2y)$. Find $\ker T$, $\operatorname{Im} T$, and state whether $T$ is invertible.
2. Find the invariant points of $T(x, y) = (5x - 4y,\ 2x - y)$.
3. Let $T(x, y) = (3x - y,\ -x + 3y)$. Find the image of the line $\ell : 2x - y = 1$.
4. Find the eigenvalues and eigenspaces of $T(x, y) = (2x + 3y,\ 3x + 2y)$. Interpret geometrically.
5. Show that if $\lambda$ is an eigenvalue of $T$, then $\lambda^2$ is an eigenvalue of $T \circ T$.

SOLUTION **Exercise 2**

1. $\det\mathbf{M} = 4 - 1 = 3 \neq 0$, so $\ker T = \{(0, 0)\}$, $\operatorname{Im} T = \mathbb{R}^2$, and $T$ is invertible with $T^{-1}(x, y) = \frac{1}{3}(2x - y,\ -x + 2y)$.
2. $T(x, y) = (x, y) \iff 4x - 4y = 0$ and $2x - 2y = 0 \iff x = y$: the invariant points form the line $y = x$.

3. Parametrise $\ell$: $(t,\ 2t-1)$. Then $T(t, 2t-1) = (3t - 2t + 1,\ -t + 6t - 3) = (t + 1,\ 5t - 3)$. With $x = t + 1, t = x - 1$, so $y = 5(x - 1) - 3 = 5x - 8$. The image is the line $5x - y = 8$.

4. $\operatorname{tr}\mathbf{M} = 4$, $\det\mathbf{M} = 4 - 9 = -5$; characteristic equation $\lambda^2 - 4\lambda - 5 = 0$, giving $\lambda_1 = 5, \lambda_2 = -1$. $E_5$: $-3x + 3y = 0 \iff y = x$, so $E_5 = \operatorname{span}\{(1,1)\}$. $E_{-1}$: $3x + 3y = 0 \iff y = -x$, so $E_{-1} = \operatorname{span}\{(1,-1)\}$. Geometrically, $T$ stretches by 5 along $y = x$ and reflects along $y = -x$: it is the reflection in $y = x$ composed with a stretch of factor 5.

5. If $T(\vec{u}) = \lambda\vec{u}$ with $\vec{u} \neq \vec{0}$, then $(T \circ T)(\vec{u}) = T(\lambda\vec{u}) = \lambda T(\vec{u}) = \lambda^2\vec{u}$, so $\lambda^2$ is an eigenvalue of $T \circ T$ with the same eigenvector. □

## 6 ACTIVITIES OF INTEGRATION, ASSESSMENT AND REMEDIATION

**EXPERIMENT** **Guided Study — The Printer's Transformation**

A printing workshop in Yaoundé uses a machine that maps every point $(x, y)$ of a design to $(x', y')$ with

$$T(x, y) = (x + 0.5y,\ 0.5y).$$

1. Show that $T$ is linear and write its matrix.
2. Compute $\det\mathbf{M}_T$. Is the machine's transformation invertible? What does this mean practically?
3. Find the invariant points of $T$.
4. The design contains the line $y = 2$. Find its image. Are horizontal lines preserved?
5. Find the eigenvalues and eigenspaces of $T$. Interpret the factor by which lengths along each eigendirection are multiplied.

Guided answers: (1) homogeneous linear components; $\mathbf{M} = \begin{pmatrix} 1 & 0.5 \\ 0 & 0.5 \end{pmatrix}$. (2) $\det\mathbf{M} = 0.5 \neq 0$: invertible, the original design can be recovered. (3) $T(x, y) = (x, y) \iff y = 0$: the $x$-axis is fixed. (4) $T(t, 2) = (t + 1, 1)$: the image is the line $y = 1$; horizontal lines stay horizontal but are shifted and compressed vertically. (5) Characteristic equation $(1 - \lambda)(0.5 - \lambda) = 0$: $\lambda_1 = 1$ with $E_1 = \operatorname{span}\{(1, 0)\}$ (horizontal lengths unchanged), $\lambda_2 = 0.5$ with $E_2$: $0.5x + 0.5y = 0$, i.e. $E_2 = \operatorname{span}\{(1, -1)\}$ (lengths along this direction halved).

**EXERCISE 3** **Assessment — GCE Style**

1. The transformation $T$ of the plane has matrix $\mathbf{M} = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix}$.

   (a) Find the image of the point $(2, -1)$.

   (b) Find the eigenvalues of $T$ and an eigenvector for each.

   (c) Deduce the equations of two lines through the origin that are invariant under $T$.

   (d) Find $T^{-1}$ and the image of the line $x + y = 4$ under $T$.

2. Let $S$ be the reflection in the line $y = \sqrt{3}\,x$.

   (a) Using $\alpha = 60^\circ$ in the reflection matrix, write the matrix of $S$.

   (b) Verify that $S \circ S = I$.

   (c) State the eigenvalues of $S$ and describe the eigenspaces geometrically.

3. $T : \mathbb{R}^3 \to \mathbb{R}^3$ is defined by $T(x, y, z) = (x - z,\ y + z,\ x + y)$.

   (a) Find $\ker T$ and $\operatorname{Im} T$; verify the rank–nullity theorem.

   (b) Is $T$ invertible? Justify.

**SOLUTION** **Assessment**

1. (a) $T(2, -1) = (6 - 1,\ 4 - 2) = (5, 2)$. (b) $\operatorname{tr}\mathbf{M} = 5$, $\det\mathbf{M} = 4$; $\lambda^2 - 5\lambda + 4 = 0$ gives $\lambda_1 = 1, \lambda_2 = 4$. $E_1$: $2x + y = 0$, eigenvector $(1, -2)$. $E_4$: $-x + y = 0$, eigenvector $(1, 1)$. (c) The invariant lines through $O$ are the eigenspaces: $y = -2x$ (points fixed) and $y = x$ (points stretched by 4). (d) $\mathbf{M}^{-1} = \frac{1}{4}\begin{pmatrix} 2 & -1 \\ -2 & 3 \end{pmatrix}$, so $T^{-1}(x, y) = \frac{1}{4}(2x - y,\ -2x + 3y)$. Line $x + y = 4$: parametrise $(t, 4 - t)$; $T(t, 4 - t) = (3t + 4 - t,\ 2t + 8 - 2t) = (2t + 4,\ 8)$. The image is the horizontal line $y = 8$.

2. (a) $\cos 120^\circ = -\frac{1}{2}$, $\sin 120^\circ = \frac{\sqrt{3}}{2}$: $\mathbf{M}_S = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$. (b) $\mathbf{M}_S^2 = \begin{pmatrix} \frac{1}{4} + \frac{3}{4} & -\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} & \frac{3}{4} + \frac{1}{4} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$. (c) Eigenvalues 1 and −1; $E_1$ is the mirror line $y = \sqrt{3}\,x$ (fixed points), $E_{-1}$ is the perpendicular line through $O$ (vectors reversed).

3. (a) Kernel: $x = z$, $y = -z$, $x + y = 0$; the third equation gives $z - z = 0$, automatically satisfied, so $\ker T = \operatorname{span}\{(1, -1, 1)\}$, nullity 1. Columns: $(1, 0, 1), (0, 1, 1), (-1, 1, 0)$; note $(-1, 1, 0) = (0, 1, 1) - (1, 0, 1)$, so $\operatorname{Im} T = \operatorname{span}\{(1, 0, 1), (0, 1, 1)\}$, a plane, rank 2. Check: $1 + 2 = 3$. □ (b) No: $\ker T \neq \{\vec{0}\}$ (equivalently $\det\mathbf{M} = 0$), so $T$ is not invertible.

**WARNING** **Remediation — Final Checklist Before the Examination**

- Always test $T(\vec{0})$ first when linearity is in doubt.
- Columns of the matrix = images of the basis vectors, in order.
- $\mathbf{M}_{S\cdot T} = \mathbf{M}_S\mathbf{M}_T$: rightmost transformation is applied first.
- Kernel: solve $\mathbf{M}\vec{u} = \vec{0}$. Image: span of the columns. Never confuse the two.
- Invariant points: solve $T(\vec{u}) = \vec{u}$, i.e. $(\mathbf{M} - \mathbf{I})\vec{u} = \vec{0}$ — this is the eigenspace $E_1$.
- Eigenvectors must be non-zero; always verify your answer by computing $T(\vec{u})$ and comparing with $\lambda\vec{u}$.
- Check eigenvalues with the trace and the determinant: $\lambda_1 + \lambda_2 = \operatorname{tr}\mathbf{M}$, $\lambda_1\lambda_2 = \det\mathbf{M}$.

# 7 KERNEL AND IMAGE OF A LINEAR TRANSFORMATION

In the previous section we discovered that a linear transformation $T$ is completely determined by its matrix $A$, and that the matrix acts on vectors by multiplication: $T(\vec{u}) = A\vec{u}$. A natural question now arises: what does a linear transformation do to the space as a whole? Does it squash the plane onto a line? Does it rotate everything faithfully? Does it send some non-zero vectors to zero?

Two sets answer these questions completely. The **kernel** tells us what is destroyed (sent to the zero vector), and the **image** tells us what is produced (the set of all outputs). Together they give a precise "identity card" of any linear transformation, and they are the key to deciding whether $T$ is injective, surjective, or invertible — which we shall need in the next section.

## 7.1 The Kernel of a Linear Transformation

**An observation to start with.**

Consider the orthogonal projection $P$ of the plane onto the $x$-axis. Every point $(x, y)$ is sent to $(x, 0)$. Now look at the points on the $y$-axis: $(0, 1)$, $(0, -3)$, $(0, 7)$ are all sent to $(0, 0)$. The whole $y$-axis is "crushed" onto the origin. This collection of vectors that $P$ annihilates is exactly what we call the kernel of $P$.

**DEFINITION** **Kernel of a linear transformation**

Let $T : E \to F$ be a linear transformation. The **kernel** of $T$, written $\ker T$, is the set of all vectors of $E$ that are sent to the zero vector of $F$:

$$\ker T = \{\, \vec{u} \in E \ : \ T(\vec{u}) = \vec{0} \,\}.$$

If $T$ has matrix $A$, then $\ker T$ is the set of solutions of the homogeneous system $AX = 0$.

Note carefully: $\ker T$ lives in the domain $E$. It is never empty, because linearity forces $T(\vec{0}) = \vec{0}$, so $\vec{0} \in \ker T$ always.

**PROPERTY** **The kernel is a subspace**

$\ker T$ is a vector subspace of $E$: it contains $\vec{0}$, and it is closed under addition and scalar multiplication.

**Proof (examinable).** We verify the three subspace conditions using linearity of $T$.

1. $T(\vec{0}) = \vec{0}$, so $\vec{0} \in \ker T$.
2. If $\vec{u}, \vec{v} \in \ker T$, then $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v}) = \vec{0} + \vec{0} = \vec{0}$, so $\vec{u} + \vec{v} \in \ker T$.
3. If $\vec{u} \in \ker T$ and $\lambda \in \mathbb{R}$, then $T(\lambda\vec{u}) = \lambda T(\vec{u}) = \lambda\vec{0} = \vec{0}$, so $\lambda\vec{u} \in \ker T$. □

This is excellent news: instead of listing infinitely many vectors, we can describe $\ker T$ by a basis. In the plane, the kernel is either the single point $\{\vec{0}\}$, a line through the origin, or the whole plane.

**METHOD** **Finding the kernel in practice**

To find $\ker T$ when $T$ has matrix $A$:

1. Write the homogeneous system $AX = 0$, where $X = \begin{pmatrix} x \\ y \end{pmatrix}$.
2. Solve it (by elimination or by inspection).
3. Describe the solution set geometrically: the origin alone, a line through the origin (give a direction vector), or a plane through the origin in 3D.

**WORKED EXAMPLE 13** **Kernel of a plane transformation**

Let $T$ be the linear transformation of the plane with matrix $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$. Determine $\ker T$ and describe it geometrically.

**SOLUTION**

Solution. We solve $AX = 0$:

$$\begin{cases} x + 2y = 0 \\ 2x + 4y = 0 \end{cases}$$

The second equation is just twice the first, so the only condition is $x = -2y$. Setting $y = t$ (a

free parameter), the solutions are

$$\begin{pmatrix} x \\ y \end{pmatrix} = t\begin{pmatrix} -2 \\ 1 \end{pmatrix}, \qquad t \in \mathbb{R}.$$

Hence $\ker T$ is the line through the origin spanned by $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$, i.e. the line $x + 2y = 0$. Every vector on this line is crushed to $\vec{0}$ by $T$.

## 7.2 The Image of a Linear Transformation

**An observation to start with.**

Return to the projection $P$ onto the $x$-axis. Whatever point $(x, y)$ of the plane we choose, its image $(x, 0)$ lies on the $x$-axis; and every point of the $x$-axis is reached (it is its own image). So the totality of outputs of $P$ is exactly the $x$-axis. This set of all outputs is the image of $P$.

**DEFINITION** **Image of a linear transformation**

Let $T : E \to F$ be a linear transformation. The **image** of $T$, written $\operatorname{Im} T$, is the set of all outputs of $T$:

$$\operatorname{Im} T = \{ T(\vec{u}) \ : \ \vec{u} \in E \} = \{ \vec{w} \in F \ : \ \vec{w} = T(\vec{u}) \text{ for some } \vec{u} \in E \}.$$

Note carefully: $\operatorname{Im} T$ lives in the codomain $F$.

**PROPERTY** **The image is a subspace**

$\operatorname{Im} T$ is a vector subspace of $F$.

**Proof (examinable).** (i) $\vec{0} = T(\vec{0}) \in \operatorname{Im} T$. (ii) If $\vec{w}_1 = T(\vec{u}_1)$ and $\vec{w}_2 = T(\vec{u}_2)$, then $\vec{w}_1 + \vec{w}_2 = T(\vec{u}_1 + \vec{u}_2) \in \operatorname{Im} T$. (iii) If $\vec{w} = T(\vec{u})$ and $\lambda \in \mathbb{R}$, then $\lambda\vec{w} = T(\lambda\vec{u}) \in \operatorname{Im} T$. □

How do we compute it concretely? Write $\vec{u} = x\vec{e}_1 + y\vec{e}_2$ in the standard basis. By linearity,

$$T(\vec{u}) = x\,T(\vec{e}_1) + y\,T(\vec{e}_2).$$

But $T(\vec{e}_1)$ and $T(\vec{e}_2)$ are precisely the **columns** of the matrix $A$! So every output is a linear combination of the columns.

**METHOD** **Finding the image in practice**

The image of $T$ is the span of the columns of its matrix $A$:

$$\operatorname{Im} T = \operatorname{span}\{T(\vec{e}_1),\ T(\vec{e}_2)\} = \operatorname{span}\{\text{columns of } A\}.$$

1. Read off the columns of $A$.
2. If they are not parallel, they span the whole plane: $\operatorname{Im} T = \mathbb{R}^2$.
3. If they are parallel (one is a multiple of the other), $\operatorname{Im} T$ is the line through the origin in that common direction.

**WORKED EXAMPLE 14** **Image of a plane transformation**

For the same matrix $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, find $\operatorname{Im} T$.

**SOLUTION**

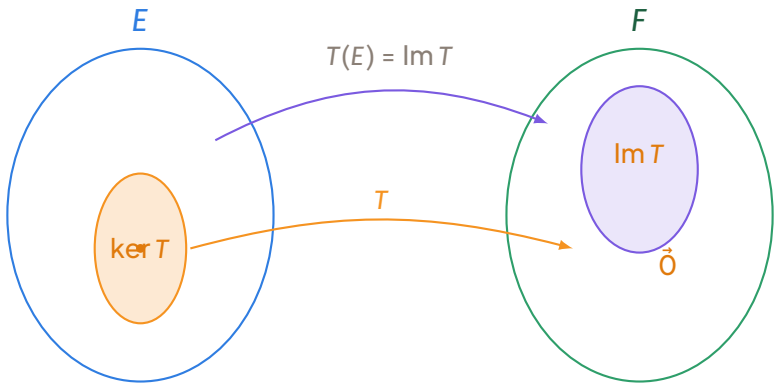
Solution. The columns are $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 4 \end{pmatrix} = 2\begin{pmatrix} 1 \\ 2 \end{pmatrix}$. They are parallel, so

$$\operatorname{Im} T = \operatorname{span}\left\{\begin{pmatrix} 1 \\ 2 \end{pmatrix}\right\},$$

the line through the origin of slope 2 (equation $y = 2x$). Notice the beautiful picture: $T$ crushes the kernel line $x + 2y = 0$ to a point, and the whole plane collapses onto the image line $y = 2x$.

**WARNING** **Do not confuse kernel and image**

The kernel is a set of inputs: it lives in the domain $E$. The image is a set of outputs: it lives in the codomain $F$. A frequent GCE error is to solve $AX = 0$ and then declare the answer to be “the image”. Remember: kernel = what is killed; image = what is reached.



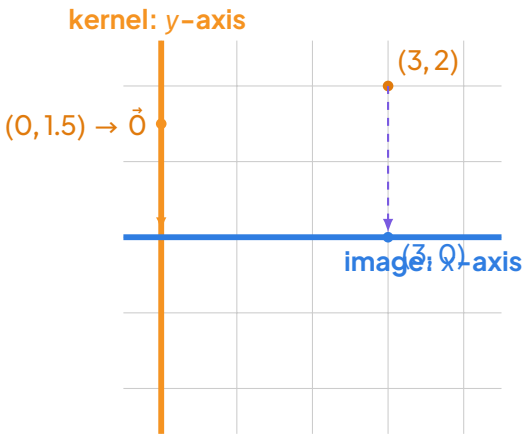

The kernel (part of the domain $E$) maps to $\vec{0}$; the image is the part of the codomain $F$ actually reached by $T$.

## 7.3 Geometric Interpretation: the Case of a Projection

Projection onto the $x$-axis is the perfect mental model. Its matrix is

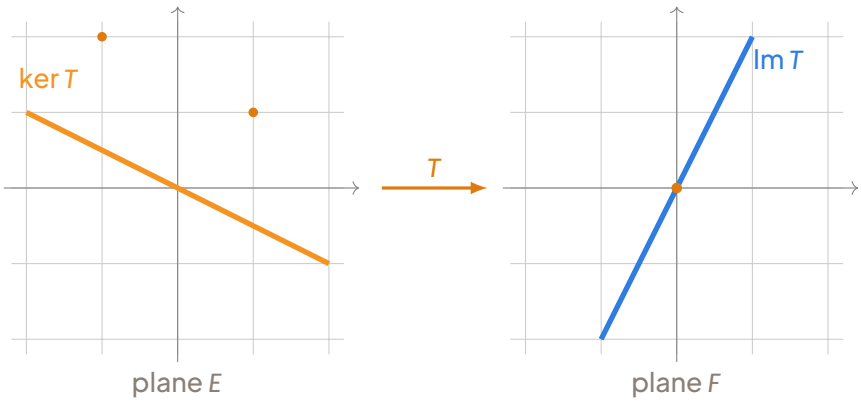
$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Solving $AX = 0$ gives $x = 0$: the kernel is the $y$-axis, the direction that is "projected away". The columns of $A$ are $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$, so the image is the $x$-axis, the line onto which we project.




Orthogonal projection onto the $x$-axis: the kernel ($y$-axis, orange) is crushed to $\vec{0}$; the image ($x$-axis, blue) receives every output.

The same phenomenon occurs for any singular $2 \times 2$ matrix: the whole plane is collapsed onto a single line (the image), and all vectors pointing in one direction (the kernel) are annihilated.




The singular transformation $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ collapses the whole plane onto the line $\operatorname{Im} T : y = 2x$; the kernel line $x + 2y = 0$ is sent to the origin.

## 7.4 Kernel, Injectivity and Determinants

When is a transformation injective (one-to-one), i.e. when do two different inputs never give the same output? Suppose $T(\vec{u}) = T(\vec{v})$. By linearity, $T(\vec{u} - \vec{v}) = \vec{0}$, so $\vec{u} - \vec{v} \in \ker T$. If the kernel contains only $\vec{0}$, we are forced to conclude $\vec{u} = \vec{v}$. This proves the fundamental criterion:

**PROPERTY** **Injectivity criterion**

A linear transformation $T$ is **injective** if and only if $\ker T = \{\vec{0}\}$.

This is a remarkable simplification: to test injectivity, we do not need to compare all pairs of vectors — we only solve one homogeneous system.

For a square matrix, solving $AX = 0$ connects directly with the determinant. The system $AX = 0$ has a unique solution (namely $X = 0$) exactly when $A$ is non-singular:

**PROPERTY** **Kernel and determinant**

Let $T$ be a linear transformation of the plane (or of space) with square matrix $A$. Then

$$\ker T = \{\vec{0}\} \iff \det A \neq 0.$$

In that case $T$ is injective, and for plane transformations it is also **surjective**: $\operatorname{Im} T$ is the whole plane, because the two columns of $A$ are then non-parallel and span $\mathbb{R}^2$.

Conversely, $\det A = 0$ means the columns are parallel: the image collapses to a line (or a point), and a whole line of vectors is sent to $\vec{0}$.

**EXEMPLE** **Quick test**

$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$: $\det A = 4 - 4 = 0$, so $\ker T$ is a line and $T$ is neither injective nor surjective — confirming our computations above. $B = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$: $\det B = 6 - 1 = 5 \neq 0$, so $\ker T = \{\vec{0}\}$ and $\operatorname{Im} T = \mathbb{R}^2$: $T$ is a bijection of the plane.

**DID YOU KNOW?** **GCE extra: nothing is lost, everything is transformed**

Here is a powerful intuition for checking your answers (stated informally, as the GCE requires): what is lost in the kernel reappears as the size of the image. In the plane: if the kernel is a line (dimension 1), the image is a line (dimension 1), and $1 + 1 = 2$, the dimension of the plane. If the kernel is just $\{\vec{0}\}$ (dimension 0), the image is the whole plane (dimension 2). In 3D, a kernel that is a plane (dimension 2) forces the image to be a line (dimension 1): $2 + 1 = 3$. This "conservation law" (the rank–nullity theorem, studied at university) lets you spot impossible answers instantly.

**WORKED EXAMPLE 15** **Kernel and image in space**

Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ have matrix $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}$. Find $\ker T$ and $\operatorname{Im} T$, and check the "conservation" of dimensions.

**SOLUTION**

Solution. Solve $AX = 0$:

$$\begin{cases} x + z = 0 \\ y + z = 0 \\ x + y + 2z = 0 \end{cases}$$

The third equation is the sum of the first two, so it adds nothing. With $z = t$: $x = -t$, $y = -t$. Hence

$$\ker T = \operatorname{span}\left\{\begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}\right\}, \quad \text{a line through the origin (dimension 1).}$$

The columns of $A$ are $\vec{c}_1 = \begin{pmatrix}1\\0\\1\end{pmatrix}$, $\vec{c}_2 = \begin{pmatrix}0\\1\\1\end{pmatrix}$, $\vec{c}_3 = \vec{c}_1 + \vec{c}_2$. Since $\vec{c}_1$ and $\vec{c}_2$ are not parallel,

$$\operatorname{Im} T = \operatorname{span}\{\vec{c}_1, \vec{c}_2\}, \quad \text{a plane through the origin (dimension 2).}$$

Check: $1 + 2 = 3 = \dim \mathbb{R}^3$. ✓

**EXERCISE 4** **Kernel and image**

For each matrix, find $\ker T$ and $\operatorname{Im} T$, state whether $T$ is injective and/or surjective, and verify the dimension check.

1. $A = \begin{pmatrix}3 & -6\\-1 & 2\end{pmatrix}$
2. $B = \begin{pmatrix}1 & 1\\0 & 1\end{pmatrix}$
3. $C = \begin{pmatrix}0 & 0\\0 & 0\end{pmatrix}$ (the zero transformation)

**SOLUTION** **Exercises on Kernel and Image**

1. $\det A = 6 - 6 = 0$. Kernel: $3x - 6y = 0 \Rightarrow x = 2y$, so $\ker T = \operatorname{span}\{(2,1)\}$, a line. Columns are parallel: $\operatorname{Im} T = \operatorname{span}\{(3,-1)\}$, a line. Neither injective nor surjective. Check: $1 + 1 = 2$. ✓
2. $\det B = 1 \neq 0$. Kernel: $x + y = 0$ and $y = 0$ force $x = y = 0$, so $\ker T = \{\vec{0}\}$. Columns $(1,0)$ and $(1,1)$ are non-parallel: $\operatorname{Im} T = \mathbb{R}^2$. $T$ is a bijection. Check: $0 + 2 = 2$. ✓
3. Every vector satisfies $CX = 0$: $\ker T = \mathbb{R}^2$ (the whole plane). The only output is $\vec{0}$: $\operatorname{Im} T = \{\vec{0}\}$. Check: $2 + 0 = 2$. ✓

**KEY POINTS** **Key points**

- $\ker T = \{\vec{u} : T(\vec{u}) = \vec{0}\}$: a subspace of the domain, found by solving $AX = 0$.
- $\operatorname{Im} T = \{T(\vec{u})\}$: a subspace of the codomain, spanned by the columns of $A$.
- $T$ injective $\iff \ker T = \{\vec{0}\} \iff \det A \neq 0$ (square matrices).

- $T$ surjective (plane) $\iff \operatorname{Im} T = \mathbb{R}^2 \iff \det A \neq 0$.
- For a projection: the kernel is the direction projected away; the image is the line (or plane) onto which we project.
- Dimension check: $\dim \ker T + \dim \operatorname{Im} T = \dim E$.

We now know exactly when a linear transformation is a bijection: when its kernel is trivial. In the next section we shall exploit this to undo the transformation — defining its inverse — and to hunt for the points that the transformation leaves fixed, the invariant points.

# 8 INVERSE OF A LINEAR TRANSFORMATION AND INVARIANT POINTS

In the previous section we saw that the kernel and the image tell us whether a linear transformation $T$ "collapses" information: if $\ker T = \{\vec{0}\}$ the transformation is injective, and if $\operatorname{Im} T$ is the whole target space it is surjective. When both hold, every output vector comes from exactly one input vector — and this means we can undo the transformation. That is the subject of this section: the **inverse** of a linear transformation, and the closely related question of which points are left unchanged by $T$, the **invariant points**.

## 8.1 Bijective Linear Transformations and the Inverse

**EXPERIMENT** **Can we go backwards?**

Consider the plane transformation $T$ with matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$. A point $P(1, 2)$ is sent to $T(P) = (4, 3)$. Now suppose someone tells you only the image $Q(4, 3)$ and asks: which point was sent here? You must solve

$$\begin{cases} 2x + y = 4 \\ x + y = 3 \end{cases}$$

which gives the unique solution $x = 1$, $y = 2$. Because $\det A = 2(1) - 1(1) = 1 \neq 0$, this system always has exactly one solution, whatever $Q$ is. The transformation can be reversed.

Contrast this with $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, where $\det A = 0$: the point $(2, 1)$ and the point $(0, 0)$ both have image $(2, 4)$, so there is no way to know where you came from. Reversibility is impossible.

**DEFINITION** **Bijective linear transformation and its inverse**

A linear transformation $T : \mathbb{R}^n \to \mathbb{R}^n$ is said to be **bijective** (or **invertible**) when it is both injective ($\ker T = \{\vec{0}\}$) and surjective ($\operatorname{Im} T = \mathbb{R}^n$). In that case every vector $\vec{v}$ has a unique pre-image, and the **inverse transformation** $T^{-1}$ is defined by

$$T^{-1}(\vec{v}) = \vec{u} \iff T(\vec{u}) = \vec{v}, \qquad \text{so that} \qquad T^{-1}\big(T(\vec{u})\big) = \vec{u} \quad \text{and} \quad T\big(T^{-1}(\vec{v})\big) = \vec{v}.$$

For plane transformations ($n = 2$), injectivity alone already implies surjectivity (by the rank–nullity theorem), so $T$ is invertible as soon as $\ker T = \{\vec{0}\}$.

**PROPERTY** **Invertibility criterion and matrix of the inverse**

Let $T$ be a linear transformation of the plane with matrix $A$. Then

$$T \text{ is invertible} \iff \det A \neq 0,$$

and in that case $T^{-1}$ is also a linear transformation, whose matrix is the inverse matrix:

$$A^{-1} = \frac{1}{\det A}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \qquad \text{for} \qquad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

The proof of the criterion (via $\ker T$ and the rank of $A$) is examinable.

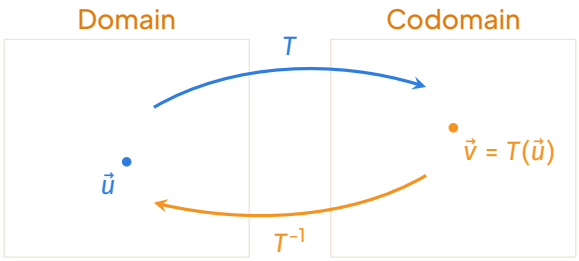
**Justification.** $T(\vec{u}) = A\vec{u} = \vec{0}$ has a non-zero solution if and only if the columns of $A$ are linearly dependent, i.e. $\det A = 0$. Hence $\ker T = \{\vec{0}\} \iff \det A \neq 0$. By the rank–nullity theorem, $\dim(\ker T)+\dim(\operatorname{Im} T) = 2$, so $\ker T = \{\vec{0}\}$ forces $\dim(\operatorname{Im} T) = 2$, i.e. $\operatorname{Im} T = \mathbb{R}^2$ (surjectivity). Linearity of $T^{-1}$ follows because for any $\vec{v}_1, \vec{v}_2$, the pre-image of $\vec{v}_1+\vec{v}_2$ under $T$ is the sum of the pre-images (since $T$ is linear). Finally $AA^{-1} = A^{-1}A = I$, so $A^{-1}(A\vec{u}) = \vec{u}$, confirming that $A^{-1}$ undoes $A$. ■

**EXEMPLE** **Computing an inverse transformation**

Let $T$ have matrix $A = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}$. Then $\det A = 3 - 2 = 1 \neq 0$, so $T$ is invertible and

$$A^{-1} = \frac{1}{1}\begin{pmatrix} 1 & -1 \\ -2 & 3 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -2 & 3 \end{pmatrix}.$$

Check: $AA^{-1} = \begin{pmatrix} 3-2 & -3+3 \\ 2-2 & -2+3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$. ✓




An invertible transformation $T$ and its inverse $T^{-1}$: each undoes the other.

## 8.2 Using the Inverse to Find Pre-images

**METHOD** **Finding the pre-image of a point**

To find the point $P$ such that $T(P) = Q$:

1. Check that $\det A \neq 0$ (otherwise $Q$ may have no pre-image, or infinitely many).
2. Compute $A^{-1}$ using the determinant formula.
3. The pre-image is $P = A^{-1}Q$.
4. Verify by computing $T(P)$ and checking it equals $Q$.

**WORKED EXAMPLE 16** **Finding a pre-image**

Let $T$ be the plane transformation with matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$. Find the point $P$ whose image is $Q(5, 3)$.

**SOLUTION**

$\det A = 2 - 1 = 1 \neq 0$, so $T$ is invertible with $A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$. Then

$$P = A^{-1}Q = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 - 3 \\ -5 + 6 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Verification: $T(2, 1) = (2 \cdot 2 + 1,\ 2 + 1) = (5, 3) = Q$. ✓ Hence $P(2, 1)$.

**WARNING** **GCE extra: check your inverse before using it**

In the exam, one arithmetic slip in $A^{-1}$ ruins every following answer. Spend ten seconds verifying $AA^{-1} = I$ before using the inverse. Also remember: $(AB)^{-1} = B^{-1}A^{-1}$ — the order reverses, exactly like putting on and taking off shoes and socks.

**DID YOU KNOW?** **Determinant of the inverse**

If $T$ is invertible, $\det(A^{-1}) = \dfrac{1}{\det A}$. This follows from $\det(AA^{-1}) = \det I = 1$. Geometrically, if $T$ scales areas by factor $|\det A|$, then $T^{-1}$ scales them by the reciprocal factor.

## 8.3 Invariant (Fixed) Points

**EXPERIMENT** **Which points do not move?**

Stand in front of a mirror. Your nose moves to a new position in the reflection — but a point on the mirror surface itself stays exactly where it is. Reflections, projections and rotations all behave differently: some points are fixed, others are not. Finding the fixed points of a transformation reveals its geometric nature.

**DEFINITION** **Invariant point**

A point $M$ with position vector $\vec{x}$ is an **invariant point** (or **fixed point**) of $T$ when

$$T(M) = M, \qquad \text{i.e.} \qquad A\vec{x} = \vec{x}.$$

Equivalently, $\vec{x}$ satisfies $(A - I)\vec{x} = \vec{0}$, where $I$ is the $2 \times 2$ identity matrix.

**METHOD** **Finding and interpreting invariant points**

1. Write the condition $A\vec{x} = \vec{x}$, i.e. $(A - I)\vec{x} = \vec{0}$.
2. Solve the resulting homogeneous system in $x$ and $y$.
3. Interpret geometrically:
   - only $(0, 0)$: the unique fixed point is the origin (e.g. a rotation about $O$);
   - a line of solutions: every point of that line is fixed (e.g. the mirror line of a reflection,

the line onto which we project);

- the whole plane: $T$ is the identity transformation.

**WARNING** **The classic trap**

Fixed points satisfy $(A - I)\vec{x} = \vec{0}$, not $A\vec{x} = \vec{0}$. Solving $A\vec{x} = \vec{0}$ gives the kernel (points sent to the origin), which is a completely different question. Subtract the identity first!

**WORKED EXAMPLE 17** **Fixed points of a reflection**

$T$ is the reflection in the line $y = x$, with matrix $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Find its invariant points.

**SOLUTION**

$A - I = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$, so $(A - I)\vec{x} = \vec{0}$ gives $-x + y = 0$, i.e. $y = x$. Every point of the mirror line is fixed — exactly as expected: points on the mirror do not move under reflection. Any point off the line, e.g. $(3, 1) \mapsto (1, 3)$, is genuinely moved.

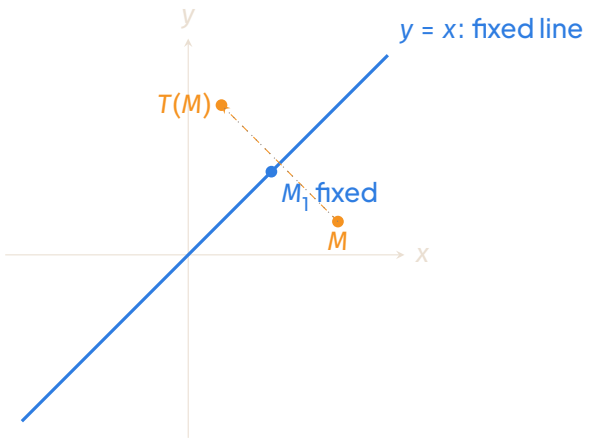
**WORKED EXAMPLE 18** **Fixed points of a projection and of a rotation**

(a) Find the fixed points of the projection onto the $x$-axis, $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. (b) Same question for the rotation of angle $\theta \neq 0$ about $O$.

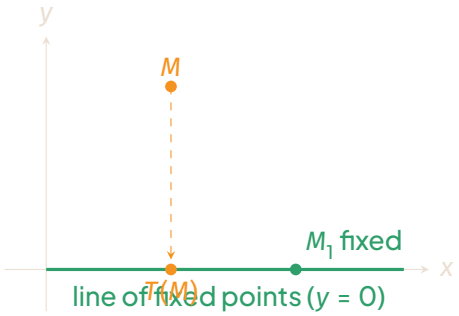
**SOLUTION**

(a) $A - I = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}$ gives $-y = 0$, i.e. $y = 0$: the fixed points are exactly the $x$-axis, the line onto which we project. Points already on the target line stay put.

(b) $A - I = \begin{pmatrix} \cos\theta - 1 & -\sin\theta \\ \sin\theta & \cos\theta - 1 \end{pmatrix}$ has determinant $(\cos\theta - 1)^2 + \sin^2\theta = 2 - 2\cos\theta \neq 0$ for $\theta \neq 0$, so the only solution is $(0, 0)$: a rotation fixes only the origin.




Reflection in $y = x$: every point of the mirror line is invariant; $M$ swaps with its image.




Projection onto the $x$-axis: the axis is the set of invariant points; $M$ slides down to its image.

## 8.4 Invariant Lines: a First Look at Eigen-directions

A line through the origin is **invariant** (as a set) when every vector on it is sent to a vector on the same line: $T(\vec{u}) = \lambda\vec{u}$ for some scalar $\lambda$. The direction of such a line is called an **eigen-direction**.

**EXEMPLE** **Invariant lines of a reflection**

For the reflection in $y = x$: the mirror line itself is invariant with $\lambda = 1$ (pointwise fixed), and the perpendicular line $y = -x$ is also invariant, since $(t, -t) \mapsto (-t, t) = -(t, -t)$, with $\lambda = -1$. These two special directions — vectors simply stretched by $T$ — are exactly the eigenvectors we shall study in Section 5, and $\lambda$ the corresponding eigenvalues.

EXPERIMENT **Discovering invariant lines**

Take the transformation $T$ with matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. Test the lines $y = x$ and $y = -x$:

- On $y = x$: vector $(t, t)$ gives $T(t, t) = (3t, 3t) = 3(t, t)$ — invariant with $\lambda = 3$.
- On $y = -x$: vector $(t, -t)$ gives $T(t, -t) = (t, -t) = 1(t, -t)$ — invariant with $\lambda = 1$.

These are the only invariant lines through the origin. This is a preview of the eigenvalue problem: find $\lambda$ and non-zero $\vec{u}$ such that $A\vec{u} = \lambda\vec{u}$.

KEY POINTS **Key points**

- $T$ is invertible $\iff \det A \neq 0$; then $T^{-1}$ is linear with matrix $A^{-1} = \dfrac{1}{\det A}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.
- Pre-image of $Q$: compute $P = A^{-1}Q$, then verify $T(P) = Q$.
- Fixed points: solve $(A - I)\vec{x} = \vec{0}$ — never $A\vec{x} = \vec{0}$.
- Geometry: mirror line (reflection), target line/plane (projection) are fixed; a rotation fixes only $O$.
- Invariant lines through $O$ give eigen-directions: $T(\vec{u}) = \lambda\vec{u}$.

EXERCISE 5 **Practice**

1. Let $T$ have matrix $A = \begin{pmatrix} 4 & 3 \\ 1 & 1 \end{pmatrix}$. Show $T$ is invertible, find $A^{-1}$, and determine the pre-image of $Q(7, 2)$.
2. Find the invariant points of the transformation with matrix $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. What do you conclude?
3. Show that the lines $y = x$ and $y = -x$ are invariant under the transformation of question 2, and find the factor $\lambda$ on each line.

**SOLUTION** **Exercise 1**

1. $\det A = 4 - 3 = 1 \neq 0$, so $T$ is invertible; $A^{-1} = \begin{pmatrix} 1 & -3 \\ -1 & 4 \end{pmatrix}$ (check: $AA^{-1} = I$). Pre-image: $P = A^{-1}Q = (7 - 6, -7 + 8) = (1, 1)$. Verify: $T(1, 1) = (7, 2)$. ✓

2. $A - I = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ gives $x + y = 0$: the fixed points form the line $y = -x$.

3. On $y = x$: $(t, t) \mapsto (3t, 3t) = 3(t, t)$, invariant with $\lambda = 3$. On $y = -x$: $(t, -t) \mapsto (t, -t)$, invariant with $\lambda = 1$ — which is why this line is pointwise fixed.

# 9 IMAGES OF LINES AND PLANES

In the previous section we studied the inverse of a linear transformation and the points left invariant by it. We now ask a geometric question of a different kind: instead of tracking a single point, we track a whole set of points. If $T$ is a linear transformation and $L$ is a line, what does the set of images $\{T(P) : P \in L\}$ look like? If $\Pi$ is a plane, what is its image? The answer is remarkably clean, and it is one of the most frequently examined ideas in this chapter: linear transformations send lines to lines (or points) and planes to planes (or lines or points). Understanding why, and being able to compute the image quickly, is the goal of this section.

## 9.1 Image of a Line Through the Origin

Let us start with the simplest case. Take a line through the origin in the plane (or in space), say the line $L$ spanned by a non-zero vector $\vec{v}$. Every point of $L$ has position vector $t\vec{v}$ for some $t \in \mathbb{R}$. Apply a linear transformation $T$:

$$T(t\vec{v}) = t\,T(\vec{v}).$$

As $t$ runs through $\mathbb{R}$, the image $t\,T(\vec{v})$ runs through the line spanned by $T(\vec{v})$ — unless $T(\vec{v}) = \vec{0}$, in which case every image is the origin. This single line of algebra, which uses nothing but the linearity of $T$, proves the following fundamental result.

**PROPERTY** **Image of a line or a plane**

Let $T$ be a linear transformation.

- The image of a **line** under $T$ is a line or a point.
- The image of a **plane** under $T$ is a plane, a line or a point.

More precisely, the line through the origin spanned by $\vec{v}$ maps onto the line spanned by $T(\vec{v})$; it collapses to the origin exactly when $\vec{v} \in \ker T$.

**WARNING** **A frequent misconception**

The image of a line that does not pass through the origin is not required to pass through the origin. Only subspaces (lines and planes through $O$) are guaranteed to map onto subspaces through $O$, because $T(\vec{0}) = \vec{0}$. A general line maps to a general line, which may or may not contain $O$. Keep this in mind whenever you read off the image line.

## 9.2 Image of a General Line: the Parametric Method

A general line $L$ is described by one point $A$ on it and a direction vector $\vec{v}$:

$$\vec{r} = \vec{a} + t\vec{v}, \qquad t \in \mathbb{R},$$

where $\vec{a}$ is the position vector of $A$. Applying $T$ and using linearity:

$$T(\vec{r}) = T(\vec{a} + t\vec{v}) = T(\vec{a}) + t\,T(\vec{v}).$$

This is again the parametric equation of a line: it passes through the point $T(A)$ and has direction vector $T(\vec{v})$. If $T(\vec{v}) = \vec{0}$, the image is the single point $T(A)$.

**METHOD** **Finding the image of a line**

To find the image of the line $\vec{r} = \vec{a} + t\vec{v}$ under the linear transformation with matrix $A$:

1. Compute the image of the point: $A\vec{a}$, giving a point $A'$ of the image line.
2. Compute the image of the direction vector: $A\vec{v}$, giving the direction $\vec{v}\,'$ of the image line.
3. Write the image line as $\vec{r}\,' = A\vec{a} + t\,(A\vec{v})$, and convert to Cartesian form if required.
4. Check the degenerate case: if $A\vec{v} = \vec{0}$, the image is the single point $A'$.

**WORKED EXAMPLE 19** **Image of a line under a shear**

The transformation $T$ has matrix $M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ (a shear parallel to the $x$-axis). Find the image of the line $L : y = x + 1$.

**SOLUTION**

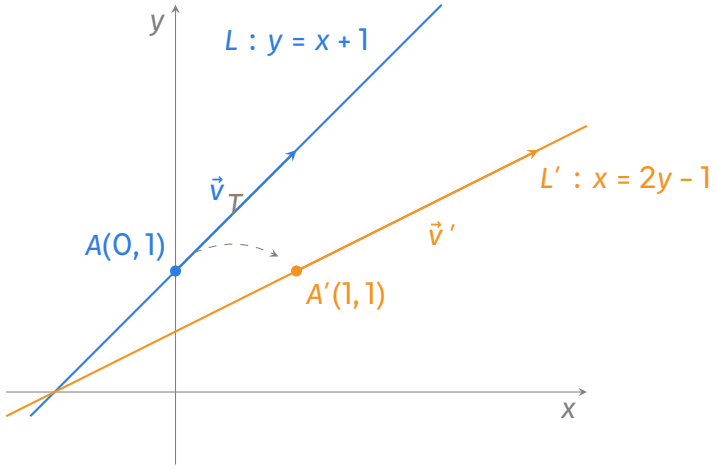
Step 1: parametrise $L$. A point on $L$ is $A(0, 1)$ and a direction vector is $\vec{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, so $\vec{r} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + t\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Step 2: transform point and direction.

$$M\vec{a} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad M\vec{v} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Step 3: write the image line. $\vec{r}\,' = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + t\begin{pmatrix} 2 \\ 1 \end{pmatrix}$, i.e. $x = 1 + 2t$, $y = 1 + t$. Eliminating $t$: $t = y - 1$, so $x = 1 + 2(y - 1) = 2y - 1$.

Conclusion: the image is the line $x = 2y - 1$, that is $y = \dfrac{x + 1}{2}$. Note that it does not pass through the origin — exactly as warned above.




The shear of matrix $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ maps $L$ to $L'$; the point $A$ and the direction vector $\vec{v}$ are tracked.

**WORKED EXAMPLE 20** **Image of a line under a rotation**

$T$ is the rotation about $O$ through $90^\circ$ anticlockwise, with matrix $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Find the image of the line $L : 2x + y = 4$.

**SOLUTION**

Parametrise: $A(2, 0)$ lies on $L$, direction $\vec{v} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$ (since $2(1) + (-2) = 0$).

$$R\vec{a} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \qquad R\vec{v} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Image line: $\vec{r}\,' = \begin{pmatrix} 0 \\ 2 \end{pmatrix} + t\begin{pmatrix} 2 \\ 1 \end{pmatrix}$, i.e. $x = 2t$, $y = 2 + t$, giving $y = 2 + \frac{x}{2}$, or $x - 2y + 4 = 0$. Geometrically this is exactly the original line rotated through $90^\circ$, as expected.

## 9.3 Image of a Plane in Space

The same idea works in three dimensions. A plane $\Pi$ through the origin is spanned by two non-parallel vectors $\vec{u}$ and $\vec{v}$: its points are $s\vec{u} + t\vec{v}$. Applying $T$:

$$T(s\vec{u} + t\vec{v}) = s\,T(\vec{u}) + t\,T(\vec{v}),$$

so the image is the set of all linear combinations of $T(\vec{u})$ and $T(\vec{v})$:

- if $T(\vec{u})$ and $T(\vec{v})$ are **not parallel**, the image is the **plane** spanned by them;
- if they are parallel (and not both zero), the image collapses to a **line**;
- if both are zero, the image is the single point $O$.

For a general plane with equation $\vec{r} = \vec{a} + s\vec{u} + t\vec{v}$, the image is $\vec{r}\,' = T(\vec{a}) + s\,T(\vec{u}) + t\,T(\vec{v})$.

**METHOD** **Finding the image of a plane**

1. Identify one point $A$ of the plane and two spanning (direction) vectors $\vec{u}, \vec{v}$.
2. Compute $T(\vec{a})$, $T(\vec{u})$, $T(\vec{v})$ using the matrix of $T$.
3. If $T(\vec{u})$ and $T(\vec{v})$ are not parallel, the image is the plane through $T(A)$ spanned by them; find its Cartesian equation using the normal vector $T(\vec{u}) \times T(\vec{v})$.
4. If they are parallel, the image is a line; if both vanish, a point.

**WORKED EXAMPLE 21** **Image of a plane in space**

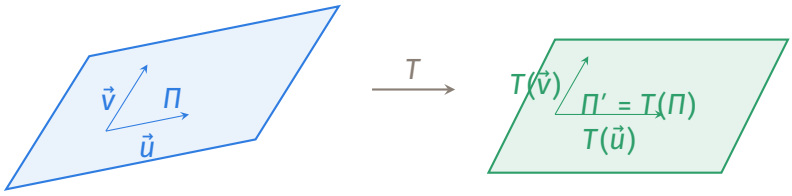
$T$ has matrix $M = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$. Find the image of the plane $\Pi : z = 0$.

**SOLUTION**

$\Pi$ passes through $O$ and is spanned by $\vec{u} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $\vec{v} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$.

$$M\vec{u} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad M\vec{v} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

These are not parallel, so the image is the plane spanned by them: again the plane $z = 0$. Note that $T$ is singular ($\det M = 0$): the whole of space is projected onto the plane $z = 0$ (since the third column is $(1, 1, 0)^T$ and the third row is zero). This particular plane $\Pi$ is invariant as a set; in fact, every point of $\Pi$ is fixed because $M(x, y, 0)^T = (x, y, 0)^T$.




A plane $\Pi$ spanned by $\vec{u}, \vec{v}$ maps to the plane $\Pi'$ spanned by $T(\vec{u}), T(\vec{v})$.

## 9.4 Singular Transformations: Collapse of Lines, Grids and Planes

When $T$ is **singular** ($\det M = 0$), its image has dimension less than the ambient space, and whole families of lines and planes collapse. In the plane, a singular transformation of rank 1 sends every line onto the same fixed line (or a point on it): the entire plane is flattened onto $\operatorname{Im} T$.

**WORKED EXAMPLE 22** **A singular transformation**

$T$ has matrix $M = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$. Find the image of the line $L : y = x + 1$, and describe the image of the whole plane.

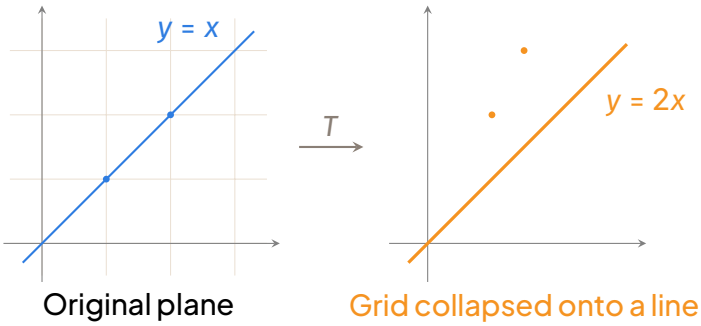
**SOLUTION**

$\det M = 4 - 4 = 0$, so $T$ is singular. Every image vector has the form

$$M\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + 2y \\ 2x + 4y \end{pmatrix} = (x + 2y)\begin{pmatrix} 1 \\ 2 \end{pmatrix},$$

so the whole plane maps onto the line $y = 2x$ (the line spanned by $(1, 2)$).

For $L$: take $A(0, 1)$, $\vec{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Then $M\vec{a} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$ and $M\vec{v} = \begin{pmatrix} 3 \\ 6 \end{pmatrix} = 3\begin{pmatrix} 1 \\ 2 \end{pmatrix}$. The image is the line through $(2, 4)$ in direction $(1, 2)$, i.e. $y = 2x$: the line $L$ maps onto the same line into which the whole plane collapses. Note $M\vec{v} \neq \vec{0}$, so $L$ does not collapse to a point — but a line with direction $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ would, since $\begin{pmatrix} 2 \\ -1 \end{pmatrix} \in \ker M$.




The singular matrix $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ collapses the whole plane (grid and line) onto the single line $y = 2x$.

In space, a singular transformation can collapse a plane onto a line: for instance with $M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, the plane $z = 0$ spanned by $(1, 0, 0)$ and $(0, 1, 0)$ maps onto the $x$-axis, because $M(1, 0, 0)^T = (1, 0, 0)^T$ but $M(0, 1, 0)^T = \vec{0}$.

## 9.5 What Is Preserved and What Is Not

Because $T(\vec{a} + t\vec{v}) = T(\vec{a}) + t\,T(\vec{v})$ with the same parameter $t$, the position of a point along a line, measured as a fraction of the way along, is unchanged.

**PROPERTY** **Preservation properties of linear transformations**

A linear transformation $T$ preserves:

- the **origin**: $T(\vec{0}) = \vec{0}$;
- **parallelism**: parallel lines map to parallel lines (or points), since a common direction $\vec{v}$ maps to the common direction $T(\vec{v})$;
- **midpoints** and **ratios of division**: if $M$ divides $AB$ in the ratio $\lambda : \mu$, then $T(M)$ divides $T(A)T(B)$ in the same ratio $\lambda : \mu$.

However, distances and angles are generally not preserved (a shear stretches lengths and distorts angles); they are preserved only by special transformations such as rotations and reflections (orthogonal matrices).

The proof of ratio preservation is one line: if $\vec{m} = \dfrac{\mu\vec{a} + \lambda\vec{b}}{\lambda + \mu}$, then by linearity $T(\vec{m}) = \dfrac{\mu\, T(\vec{a}) + \lambda\, T(\vec{b})}{\lambda + \mu}$, which is exactly the point dividing $T(A)T(B)$ in the ratio $\lambda : \mu$.

DID YOU KNOW? **Computer Graphics and Linear Transformations**

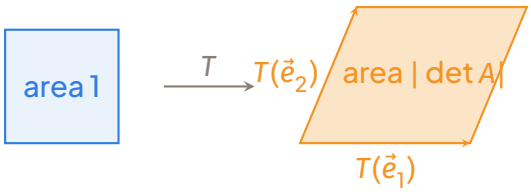
In 3D computer graphics, every object is represented by a mesh of vertices (points). To move, rotate, scale, or shear an object, the graphics pipeline applies a linear transformation (represented by a $4 \times 4$ matrix in homogeneous coordinates) to every vertex. The fact that linear transformations map straight lines to straight lines means that the edges of the mesh (line segments between vertices) remain straight — we only need to transform the endpoints! This property makes real-time rendering of complex scenes computationally feasible.

## 9.6 GCE Extra: Area and Volume Scaling

PROPERTY **Area and volume scaling**

If $T$ has matrix $A$, then areas in the plane are multiplied by $|\det A|$, and volumes in space are multiplied by $|\det A|$. In particular, the unit square maps to a parallelogram of area $|\det A|$. This is a quick check on your work and a favourite one-mark question.

For the shear $M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $\det M = 1$: areas are preserved. For the singular matrix $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, $\det = 0$: every region collapses to zero area, confirming the flattening we saw above.




The unit square maps to the parallelogram spanned by the columns of $A$; its area is $|\det A|$.

**EXERCISE 6** **GCE practice**

1. $T$ has matrix $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$. Find the image of the line $x + y = 2$, and state the factor by which areas are scaled.
2. $T$ has matrix $\begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix}$. Show that the whole plane maps onto a line, find that line, and find the image of the line $y = x$.
3. In space, $T$ has matrix $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$. Find the image of the plane $x + y + z = 1$.

**SOLUTION** **GCE practice**

1. Point $A(2, 0)$, direction $\vec{v} = (1, -1)^T$. Images: $A\vec{a} = (4, 0)^T$, $A\vec{v} = (1, -3)^T$. Image line: $\vec{r} = (4, 0)^T + t(1, -3)^T$, i.e. $3x + y = 12$. Areas scale by $|\det| = 6$.
2. Every image is $(x - y)(1, 2)^T$, so the plane maps onto the line $y = 2x$. For $y = x$: direction $(1, 1)^T$ maps to $\vec{0}$, so the whole line maps to the single point $O$ (indeed $y = x$ is the kernel direction, and the line passes through $O$).
3. The plane has point $A(1, 0, 0)$ and spanning vectors $\vec{u} = (1, -1, 0)^T$, $\vec{v} = (1, 0, -1)^T$. Images: $M\vec{a} = (1, 0, 1)^T$, $M\vec{u} = (1, -1, 0)^T$, $M\vec{v} = (1, 0, -1)^T$. Normal: $(1, -1, 0) \times (1, 0, -1) = (1, 1, 1)$. Image plane: $x + y + z = 1 + 0 + 1 = 2$, i.e. $x + y + z = 2$.

**KEY POINTS** **Key points**

- Image of $\vec{r} = \vec{a} + t\vec{v}$: transform the point and the direction separately; the image is $\vec{r}\,' = T(\vec{a}) + t\,T(\vec{v})$.

- A line maps to a line or a point; a plane maps to a plane, a line or a point.
- Lines through $O$ map to lines through $O$; general lines need not.
- Parallelism, midpoints and division ratios are preserved; distances and angles are not.
- Areas scale by $|\det A|$; $\det A = 0$ means total collapse of area.

# 10 EIGENVALUES AND EIGENVECTORS

In the previous section we studied the images of lines and planes under a linear transformation $T$. We saw that a generic line through the origin is sent to a different line through the origin: the transformation rotates and stretches most directions. But a natural question arises: are there special directions that $T$ leaves invariant — directions along which a vector is simply stretched, compressed or reversed, but never rotated away? These special directions, when they exist, are given by the **eigenvectors** of $T$, and the stretch factors are the **eigenvalues**. They are the key to understanding the deep structure of a linear transformation, and they appear every year in the GCE Advanced Level paper.

## 10.1 Intuition and Definition

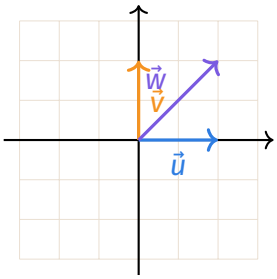
**EXPERIMENT** **Looking for invariant directions**

Consider the linear transformation of the plane given by

$$A = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}.$$

Take the vector $\vec{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Then $A\vec{u} = \begin{pmatrix} 2 \\ 0 \end{pmatrix} = 2\vec{u}$: the vector keeps its direction and is doubled.

Take $\vec{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Then $A\vec{v} = \begin{pmatrix} 0 \\ 1/2 \end{pmatrix} = \frac{1}{2}\vec{v}$: same direction, halved length. But take $\vec{w} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$: $A\vec{w} = \begin{pmatrix} 2 \\ 1/2 \end{pmatrix}$, which is not a multiple of $\vec{w}$ — the direction has changed. The two coordinate axes are invariant directions; the diagonal is not.




Original grid

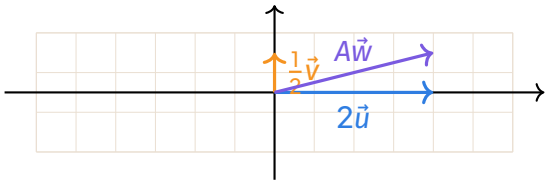



Image under $A$: horizontal stretch ×2, vertical compression ×$\frac{1}{2}$

The two eigen-directions of $A = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}$: the horizontal axis is stretched by 2, the vertical axis compressed by $\frac{1}{2}$. A generic vector $\vec{w}$ changes direction.

**DEFINITION** **Eigenvalue and eigenvector**

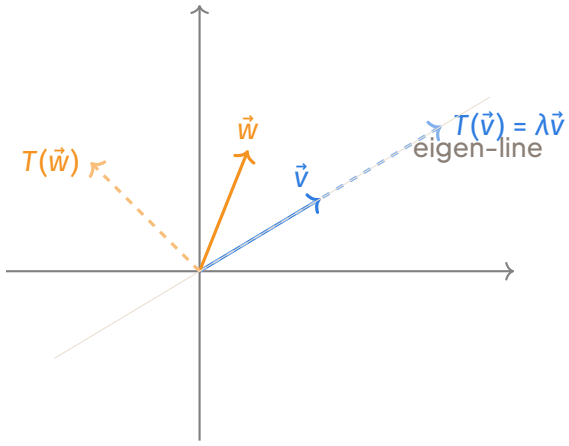
Let $T$ be a linear transformation with matrix $A$. A non-zero vector $\vec{v}$ is called an **eigenvector** of $T$ (or of $A$) if there exists a scalar $\lambda$ such that

$$T(\vec{v}) = A\vec{v} = \lambda\vec{v}.$$

The scalar $\lambda$ is called the **eigenvalue** corresponding to $\vec{v}$. The set of all solutions of $A\vec{v} = \lambda\vec{v}$ (including $\vec{0}$) is a subspace called the **eigenspace** of $\lambda$.

Geometrically, an eigenvector spans an **invariant line through the origin**: every vector on that line stays on that line after transformation. Along it, $T$ acts as a simple scaling:

- $\lambda > 1$: a **stretch**;
- $0 < \lambda < 1$: a **compression**;
- $\lambda < 0$: a stretch combined with a **reversal** of sense;
- $\lambda = 1$: the line is fixed pointwise;
- $\lambda = 0$: the direction is annihilated — the vector is sent to $\vec{0}$.




An eigenvector $\vec{v}$ keeps its line (its image is collinear with it); a non-eigenvector $\vec{w}$ changes direction.

**WARNING** **The zero vector is never an eigenvector**

The equation $A\vec{v} = \lambda\vec{v}$ is satisfied by $\vec{v} = \vec{0}$ for every $\lambda$, which would make the notion meaningless. By definition, an eigenvector must be non-zero. Saying "$\vec{0}$ is an eigenvector" is a classic exam error.

## 10.2 The Characteristic Equation

How do we find the eigenvalues of a given matrix $A$? Rewrite the defining equation:

$$A\vec{v} = \lambda\vec{v} \iff A\vec{v} - \lambda\vec{v} = \vec{0} \iff (A - \lambda I)\vec{v} = \vec{0},$$

where $I$ is the identity matrix. This is a homogeneous system. Since we demand $\vec{v} \neq \vec{0}$, the system must have non-trivial solutions, which happens exactly when the matrix $A - \lambda I$ is **singular**:

$$\det(A - \lambda I) = 0.$$

**KEY POINTS** **Characteristic equation**

The eigenvalues of $A$ are the solutions of the **characteristic equation**

$$\det(A - \lambda I) = 0.$$

For a $2 \times 2$ matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ this is the quadratic

$$\lambda^2 - (a + d)\lambda + (ad - bc) = 0, \qquad \text{i.e.} \qquad \lambda^2 - (\operatorname{tr} A)\lambda + \det A = 0.$$

**WARNING** **Subtract $\lambda$ on the diagonal only**

A very common sign error: students write $A - \lambda I$ incorrectly. Remember that $\lambda I = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$, so you subtract $\lambda$ only from the diagonal entries:

$$A - \lambda I = \begin{pmatrix} a - \lambda & b \\ c & d - \lambda \end{pmatrix}.$$

Subtracting $\lambda$ from every entry, or adding it, destroys the whole computation.

**METHOD** **Step 1 — Find the eigenvalues**

1. Form the matrix $A - \lambda I$ (subtract $\lambda$ on the diagonal only).
2. Compute $\det(A - \lambda I)$ and expand: you obtain the characteristic polynomial.
3. Solve $\det(A - \lambda I) = 0$. For $2 \times 2$, this is a quadratic; for $3 \times 3$, a cubic (look for an integer root by trial).

**METHOD** **Step 2 — Find the eigenvectors (eigenspaces)**

For each eigenvalue $\lambda$ found in Step 1:

1. Substitute $\lambda$ into $A - \lambda I$.
2. Solve the homogeneous system $(A - \lambda I)X = 0$. The two equations are always proportional (check this — it confirms your eigenvalue!); keep one.
3. Write the general solution: it is a line through the origin, e.g. $\begin{pmatrix} x \\ y \end{pmatrix} = t\begin{pmatrix} p \\ q \end{pmatrix}$, $t \in \mathbb{R}$. Any non-zero multiple of $\begin{pmatrix} p \\ q \end{pmatrix}$ is an eigenvector.

## 10.3 Worked Examples for $2 \times 2$ Matrices

**WORKED EXAMPLE 23** **Two distinct real eigenvalues**

Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

**SOLUTION**

Step 1. $A - \lambda I = \begin{pmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix}$, so

$$\det(A - \lambda I) = (4-\lambda)(3-\lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 2)(\lambda - 5).$$

Eigenvalues: $\lambda_1 = 2, \lambda_2 = 5$.

Step 2. For $\lambda_1 = 2$: $(A - 2I)X = 0$ gives $\begin{cases} 2x + y = 0 \\ 2x + y = 0 \end{cases}$, so $y = -2x$: eigenvectors $t\begin{pmatrix} 1 \\ -2 \end{pmatrix}$.

For $\lambda_2 = 5$: $(A - 5I)X = 0$ gives $-x + y = 0$, so $y = x$: eigenvectors $t\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Check: $A\begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \end{pmatrix} = 2\begin{pmatrix} 1 \\ -2 \end{pmatrix}$ ✓; $A\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix} = 5\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ ✓.

**WORKED EXAMPLE 24** **A repeated eigenvalue**

Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$.

**SOLUTION**

$\det(A - \lambda I) = (3-\lambda)^2 = 0$, so $\lambda = 3$ is a double eigenvalue (algebraic multiplicity 2).

$(A - 3I)X = 0$ gives $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, i.e. $y = 0$. The eigenspace is only the line $t\begin{pmatrix} 1 \\ 0 \end{pmatrix}$: there is only one invariant direction (geometric multiplicity 1), even though the eigenvalue is repeated. Such a matrix (a shear-like map) cannot be diagonalised.

**WORKED EXAMPLE 25** **No real eigenvalues: a rotation**

Find the eigenvalues of the rotation matrix $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ (rotation through $90^\circ$ about $O$).

**SOLUTION**

$\det(A - \lambda I) = \lambda^2 + 1 = 0$, which has no real solution. This makes perfect geometric sense: a 90° rotation moves every non-zero vector off its own line, so no real invariant direction exists. (Over $\mathbb{C}$ the eigenvalues are $\lambda = \pm i$.)

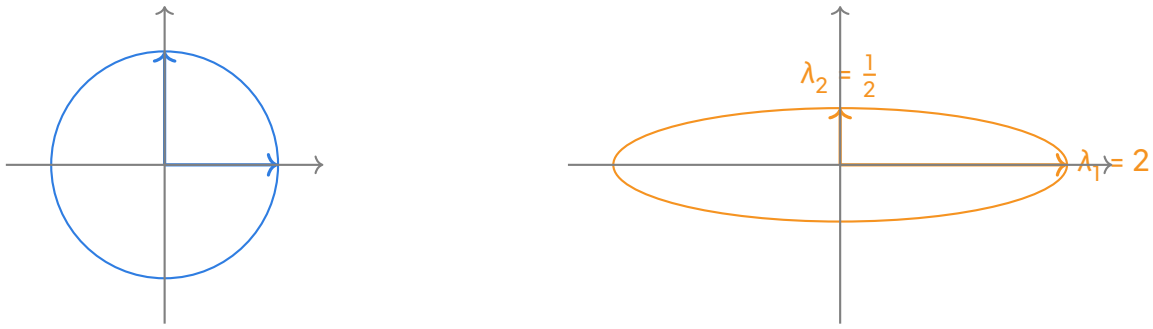
## 10.4 Links with Kernel, Image and Invariant Points

**PROPERTY** **Eigenvalue 1 and eigenvalue 0**

Let $T$ be a linear transformation with matrix $A$.

- The eigenvectors with eigenvalue $\lambda = 1$ are exactly the non-zero invariant (fixed) points of $T$: $T(\vec{v}) = \vec{v}$. The eigenspace for $\lambda = 1$ is the set of fixed points of Section 3.
- The eigenvectors with eigenvalue $\lambda = 0$ are exactly the non-zero vectors of the kernel: $T(\vec{v}) = \vec{0}$. Hence $\ker T$ is the eigenspace for $\lambda = 0$, and $T$ is singular (non-invertible) if and only if 0 is an eigenvalue.

This unifies the whole chapter: fixed points, kernel, and invariant lines are all special cases of the single equation $A\vec{v} = \lambda\vec{v}$.




Unit circle, eigen-directions → Image ellipse, axes along eigen-directions

Summary: a linear transformation maps the unit circle to an ellipse whose principal axes lie along the eigen-directions, with semi-axis lengths $|\lambda_1|$ and $|\lambda_2|$.

## 10.5 Extension to 3 × 3 Matrices

For a 3 × 3 matrix the characteristic equation $\det(A - \lambda I) = 0$ is a **cubic**. At GCE level the matrices are chosen so that the cubic has simple integer roots: expand the determinant along a row or column with zeros, then look for an integer root $\lambda_0$ by trial ($\pm 1, \pm 2, \pm 3, \ldots$), factor out $(\lambda - \lambda_0)$, and solve the remaining quadratic.

**WORKED EXAMPLE 26** **A 3 × 3 example**

Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$.

**SOLUTION**

Expanding along the third row:

$$\det(A - \lambda I) = (3 - \lambda)\big[(2 - \lambda)^2 - 1\big] = (3 - \lambda)(\lambda^2 - 4\lambda + 3) = (3 - \lambda)^2(1 - \lambda).$$

Eigenvalues: $\lambda = 1$ and $\lambda = 3$ (double, algebraic multiplicity 2).

For $\lambda = 1$: $(A - I)X = 0$ gives $x + y = 0$ and $2z = 0$, so eigenvectors $t\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$ (geometric multiplicity 1).

For $\lambda = 3$: $(A - 3I)X = 0$ gives $-x + y = 0$, $z$ free, so eigenvectors $\begin{pmatrix} x \\ x \\ z \end{pmatrix} = s\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$: here the eigenspace is a plane through the origin (geometric multiplicity 2).

## 10.6 GCE Extra: Powers of a Matrix by Diagonalisation

**(Beyond the core syllabus — a recurring advanced question type.)** Suppose a $2 \times 2$ matrix $A$ has two distinct eigenvalues $\lambda_1, \lambda_2$ with eigenvectors $\vec{v}_1, \vec{v}_2$. Form the matrix $P = (\vec{v}_1\ \vec{v}_2)$ whose columns are the eigenvectors. Then

$$P^{-1}AP = D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \qquad \text{so} \qquad A = PDP^{-1}.$$

The power of diagonalisation: $A^2 = (PDP^{-1})(PDP^{-1}) = PD^2P^{-1}$, and in general

$$A^n = PD^nP^{-1} = P\begin{pmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{pmatrix}P^{-1},$$

because $P^{-1}P = I$ cancels at every junction. This turns the hard problem of computing $A^n$ into trivial powers of scalars.

**WORKED EXAMPLE 27** **Computing $A^n$**

Given $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$, express $A^n$ in closed form.

**SOLUTION**

From the first worked example: $\lambda_1 = 2, \vec{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$; $\lambda_2 = 5, \vec{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Take $P = \begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix}$; then $\det P = 3$ and $P^{-1} = \frac{1}{3}\begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}$. Hence

$$A^n = P\begin{pmatrix} 2^n & 0 \\ 0 & 5^n \end{pmatrix}P^{-1} = \frac{1}{3}\begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix}\begin{pmatrix} 2^n & 0 \\ 0 & 5^n \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}$$

$$= \frac{1}{3}\begin{pmatrix} 2^n + 2\cdot 5^n & -2^n + 5^n \\ -2^{n+1} + 2\cdot 5^n & 2^{n+1} + 5^n \end{pmatrix}.$$

Check for $n = 1$: $\frac{1}{3}\begin{pmatrix} 2+10 & -2+5 \\ -4+10 & 4+5 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} = A$ ✓.

**WARNING** **Pitfalls in diagonalisation**

- The columns of $P$ must be eigenvectors in the same order as the eigenvalues in $D$.
- Diagonalisation requires enough independent eigenvectors: the repeated-eigenvalue shear $\begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$ above cannot be diagonalised.
- Do not forget the factor $\frac{1}{\det P}$ when computing $P^{-1}$.

**DID YOU KNOW?** **Why “eigen”?**

The prefix comes from the German word eigen, meaning “own” or “characteristic”: eigenvectors are the transformation’s “own” directions. Eigenvalues are everywhere in modern science: they describe the natural vibration frequencies of bridges and engines, the energy levels of atoms in quantum mechanics, and they are at the heart of the PageRank algorithm that orders web search results.

**EXERCISE 7** **Practice**

1. Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} 5 & -2 \\ 4 & -1 \end{pmatrix}$. Describe geometrically the action of $A$ along each eigen-direction.
2. Show that $A = \begin{pmatrix} 2 & 2 \\ 1 & 3 \end{pmatrix}$ has eigenvalues $1$ and $4$, and compute $A^n$ by diagonalisation.
3. Find the eigenvalues of $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix}$ and an eigenvector for each.
4. Explain why a reflection in the line $y = x$ has eigenvalues $1$ and $-1$, and identify the corresponding eigenvectors.

**SOLUTION** **Exercise 1**

$\det(A - \lambda I) = (5 - \lambda)(-1 - \lambda) + 8 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3)$. Eigenvalues $1$ and $3$. For $\lambda = 1$: $4x - 2y = 0$, eigenvectors $t\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ — a fixed line. For $\lambda = 3$: $2x - 2y = 0$, eigenvectors $t\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ — a direction stretched by $3$.

**SOLUTION** **Exercise 2**

$\det(A - \lambda I) = (2 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 5\lambda + 4 = (\lambda - 1)(\lambda - 4)$. Eigenvectors: for $\lambda = 1$, $t\begin{pmatrix} 2 \\ -1 \end{pmatrix}$; for $\lambda = 4$, $t\begin{pmatrix} 1 \\ 1 \end{pmatrix}$. With $P = \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}$, $P^{-1} = \frac{1}{3}\begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}$:

$$A^n = \frac{1}{3}\begin{pmatrix} 2 + 4^n & -2 + 2 \cdot 4^n \\ -1 + 4^n & 1 + 2 \cdot 4^n \end{pmatrix}.$$

**SOLUTION** **Exercise 3**

The matrix is block-triangular; expanding along the first column, $\det(A - \lambda I) = (1 - \lambda)^2(2 - \lambda)$. Eigenvalues $\lambda = 1$ (double, algebraic multiplicity $2$) and $\lambda = 2$. For $\lambda = 1$: $(A - I)X = 0$ gives

$y = 0$, so $x$ and $z$ are free. The eigenspace is the plane spanned by $\begin{pmatrix}1\\0\\0\end{pmatrix}$ and $\begin{pmatrix}0\\0\\1\end{pmatrix}$ (geometric multiplicity 2). For $\lambda = 2$: $-x + y = 0$ and $y - z = 0$, eigenvectors $t\begin{pmatrix}1\\1\\1\end{pmatrix}$.

SOLUTION **Exercise 4**

Reflection in $y = x$ fixes every vector on the line $y = x$, so $\begin{pmatrix}1\\1\end{pmatrix}$ is an eigenvector with $\lambda = 1$. It reverses every vector perpendicular to that line, so $\begin{pmatrix}1\\-1\end{pmatrix}$ is an eigenvector with $\lambda = -1$. Its matrix $\begin{pmatrix}0 & 1\\1 & 0\end{pmatrix}$ confirms: $\det(A - \lambda I) = \lambda^2 - 1 = 0$.

# 11 INTEGRATION ACTIVITY

## 11.1 Aims of the activity

This integration activity mobilises, in a single authentic modelling task, all the skills developed in this chapter:

- recognise a linear transformation and write its **matrix** in the standard basis;
- verify the **linearity** axioms $T(\vec{u}+\vec{v}) = T(\vec{u}) + T(\vec{v})$ and $T(\lambda\vec{u}) = \lambda T(\vec{u})$;
- determine the **kernel** and the **image** of a linear transformation and interpret them geometrically;
- decide whether a transformation is **bijective** and compute its **inverse**;
- find the **invariant points** of a transformation;
- compute the **image of a line** under a linear transformation;
- compute **eigenvalues** and **eigenvectors** and interpret them as privileged directions of stretching;
- relate the **determinant** of the matrix to the **area scale factor** of the map.

Beyond the technical skills, the activity develops modelling, teamwork and scientific communication: each group must present its conclusions on a poster, with a diagram, as a real cartography team would.

## 11.2 Situation

**The Cartographer of Yaoundé: Designing a Map Projection.**

A student in the Department of Geography of the University of Yaoundé I is preparing a simplified digital map of the city centre. She takes the Poste Centrale as the origin $O$ of a coordinate system, the unit on each axis being one kilometre, the $x$-axis pointing East and the $y$-axis pointing North. A point of the city is then represented by a vector $\begin{pmatrix} x \\ y \end{pmatrix}$.

To produce the map on a screen, the computer applies to every point of the city the transformation

$$T(x, y) = (2x + y,\ x + 2y).$$

In other words, the point with city coordinates $(x, y)$ is drawn on the screen at the point with map coordinates $(x', y')$ where

$$x' = 2x + y, \qquad y' = x + 2y.$$

The student must understand this transformation completely before trusting the map:

- Is $T$ genuinely a linear transformation, so that the whole theory of this chapter applies?
- Can two different streets of the city collapse onto the same point of the map (which would make the map ambiguous)? Equivalently, is $T$ bijective, and can one recover the city coordinates from the map coordinates?

- Which points of the city, if any, are drawn exactly where they are (invariant points)?
- The main avenue is modelled by the straight line of equation $y = x + 1$. What does it become on the map?
- Are there privileged directions of the city which are simply stretched by the projection, without being rotated? By what factor?
- The map will be used to estimate land areas. By what factor are areas multiplied?

Working in groups of three or four, you play the role of the cartography team. Answer the tasks below, then prepare a poster summarising your findings, including a diagram showing the eigendirections and the image of the avenue.

## 11.3 Tasks

**Task 1 — Matrix and linearity.**

1. Write the matrix $A$ of $T$ in the standard basis $\left(\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}\right)$.
2. Prove that $T$ is linear: for all vectors $\vec{u} = (x_1, y_1)$, $\vec{v} = (x_2, y_2)$ and every scalar $\lambda$, show that $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$ and $T(\lambda\vec{u}) = \lambda T(\vec{u})$.

**Task 2 — Kernel, image and bijectivity.**

1. Determine $\ker T$, the set of points of the city sent to the origin of the map.
2. Deduce that $T$ is injective, then determine $\operatorname{Im} T$ and conclude that $T$ is bijective. Interpret: can two distinct points of the city have the same image on the map?
3. Compute $\det A$ and verify that $\det A \neq 0$, confirming bijectivity.

**Task 3 — The inverse transformation.**

1. Compute $A^{-1}$.
2. Deduce the formula giving the city coordinates $(x, y)$ from the map coordinates $(x', y')$.
3. A landmark appears on the map at $(x', y') = (4, 5)$. Where is it in the city?

**Task 4 — Invariant points.**

Determine all points $(x, y)$ such that $T(x, y) = (x, y)$. Interpret the result for the map.

**Task 5 — Image of the avenue.**

The avenue is the line $D$ of equation $y = x + 1$.

1. Parametrise $D$ and compute the image of a general point of $D$.
2. Show that the image is a straight line and give its Cartesian equation in the coordinates $(x', y')$.

**Task 6 — Eigenvalues and eigenvectors.**

1. Compute the characteristic polynomial of $A$ and deduce the eigenvalues $\lambda_1$ and $\lambda_2$.
2. For each eigenvalue, find the corresponding eigenspace.

3. Interpret: which directions of the city are simply stretched by the projection, and by what factor? Draw a diagram showing the two eigen-directions.

**Task 7 — Area scale factor.**

Using $\det A$, state by how much areas are multiplied on the map. A plot of land of area 2 km$^2$ in the city occupies what area on the map?

**Task 8 — Poster.**

Prepare a group poster summarising all results, with a TikZ-style diagram of the eigen-directions and of the image of the avenue.

## 11.4 Model answer

**Task 1.** Since $T(x, y) = (2x + y,\ x + 2y)$, the matrix of $T$ in the standard basis is

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \text{so that} \quad \begin{pmatrix} x' \\ y' \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix}.$$

Linearity. Let $\vec{u} = (x_1, y_1)$, $\vec{v} = (x_2, y_2)$ and $\lambda \in \mathbb{R}$.

$$\begin{aligned} T(\vec{u} + \vec{v}) &= T(x_1 + x_2,\ y_1 + y_2) \\ &= \big(2(x_1 + x_2) + (y_1 + y_2),\ (x_1 + x_2) + 2(y_1 + y_2)\big) \\ &= (2x_1 + y_1,\ x_1 + 2y_1) + (2x_2 + y_2,\ x_2 + 2y_2) = T(\vec{u}) + T(\vec{v}). \end{aligned}$$

Similarly,

$$T(\lambda\vec{u}) = (2\lambda x_1 + \lambda y_1,\ \lambda x_1 + 2\lambda y_1) = \lambda(2x_1 + y_1,\ x_1 + 2y_1) = \lambda T(\vec{u}).$$

Hence $T$ is a linear transformation. (One could also quote the theorem: every transformation given by a matrix is linear.)

**Task 2.** The kernel is the set of $(x, y)$ with $T(x, y) = (0, 0)$:

$$\begin{cases} 2x + y = 0 \\ x + 2y = 0 \end{cases}$$

From the first equation $y = -2x$; substituting into the second, $x - 4x = 0$, so $x = 0$ and $y = 0$. Therefore

$$\ker T = \{(0, 0)\}.$$

Since the kernel is reduced to the zero vector, $T$ is **injective**. By the rank–nullity theorem, $\dim \operatorname{Im} T = 2 - 0 = 2$, so $\operatorname{Im} T = \mathbb{R}^2$: $T$ is **surjective**, hence **bijective**. Also $\det A = 2 \times 2 - 1 \times 1 = 3 \neq 0$, which confirms bijectivity. Interpretation: no two distinct points of the city ever collapse onto the same point of the map, and every point of the map corresponds to a real point of the city.

**Task 3.** Using the formula for the inverse of a $2 \times 2$ matrix,

$$A^{-1} = \frac{1}{\det A} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

Hence the inverse transformation is

$$T^{-1}(x', y') = \left( \frac{2x' - y'}{3},\ \frac{-x' + 2y'}{3} \right).$$

For the landmark at $(x', y') = (4, 5)$:

$$x = \frac{2(4) - 5}{3} = 1, \qquad y = \frac{-4 + 2(5)}{3} = 2.$$

The landmark is at $(1, 2)$ in the city, i.e. 1 km East and 2 km North of the Poste Centrale. Check: $T(1, 2) = (2 + 2,\ 1 + 4) = (4, 5)$. ✓

**Task 4.** Invariant points satisfy $T(x, y) = (x, y)$:

$$\begin{cases} 2x + y = x \\ x + 2y = y \end{cases} \iff \begin{cases} x + y = 0 \\ x + y = 0 \end{cases} \iff y = -x.$$

The set of invariant points is the line $y = -x$. Interpretation: every point of the city lying on the South-East/North-West diagonal through the Poste Centrale is drawn exactly at its true position on the map.

**Task 5.** Parametrise the avenue: a point of $D$ is $(t,\ t + 1)$, $t \in \mathbb{R}$. Its image is

$$T(t, t + 1) = \big(2t + (t + 1),\ t + 2(t + 1)\big) = (3t + 1,\ 3t + 2).$$

So $x' = 3t + 1$ and $y' = 3t + 2$. Eliminating $t$: $t = \dfrac{x' - 1}{3}$, hence

$$y' = 3 \cdot \frac{x' - 1}{3} + 2 = x' + 1.$$

The image of the avenue is the straight line $y' = x' + 1$. Remarkably, this line is globally invariant: the avenue is mapped onto itself (though each of its points moves along it, except the invariant point where $D$ meets $y = -x$, namely $\left(-\frac{1}{2}, \frac{1}{2}\right)$).

**Task 6.** The characteristic polynomial is

$$\det(A - \lambda I) = \begin{vmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{vmatrix} = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3).$$

The eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 3$.

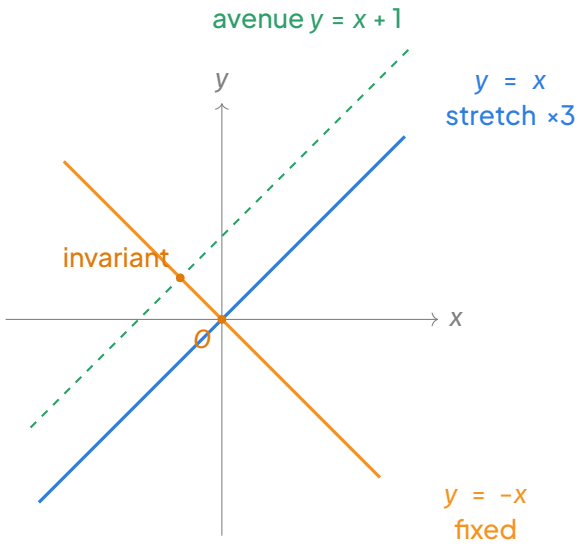
For $\lambda_1 = 1$: $(A - I)\vec{v} = \vec{0}$ gives $x + y = 0$, so the eigenspace is

$$E_1 = \operatorname{span}\begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \text{the line } y = -x.$$

For $\lambda_2 = 3$: $(A - 3I)\vec{v} = \vec{0}$ gives $-x + y = 0$, so the eigenspace is

$$E_3 = \operatorname{span}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \text{the line } y = x.$$

Interpretation: the direction $y = -x$ (the diagonal through the Poste Centrale) is left fixed (stretch factor 1) — this is exactly the line of invariant points found in Task 4. The perpendicular direction $y = x$ is stretched by a factor 3, without rotation. Every other direction is a mixture of these two effects.




Eigen-directions of the projection and image of the avenue.

**Task 7.** A linear transformation multiplies all areas by $|\det A| = 3$. Hence every area on the map is 3 times the true area. The plot of 2 km$^2$ occupies $3 \times 2 = 6$ km$^2$ on the map. Note the coherence with Task 6: the area factor equals the product of the eigenvalues, $1 \times 3 = 3$.

**Task 8.** The poster should display: the matrix $A$ and its inverse; $\ker T = \{\vec{0}\}$ and $\operatorname{Im} T = \mathbb{R}^2$ with their interpretation; the invariant line $y = -x$; the image line $y' = x' + 1$; the eigenvalues 1 and 3 with their eigen-directions; the area factor 3; and the diagram above.

**WARNING** **Pitfalls to avoid in this activity**

- Do not confuse the invariant points (solutions of $T(\vec{v}) = \vec{v}$, eigenvalue 1) with a globally invariant line (a line mapped onto itself, like the avenue here).
- The area factor is $|\det A|$, not $\det A$ in general — here $\det A = 3 > 0$ so both coincide.
- When eliminating the parameter in Task 5, keep the coordinates $(x', y')$ of the image plane; do not mix them with the city coordinates.

# 12 METHODS AND WORKED EXAMPLES

# 13 LINEAR TRANSFORMATIONS AND THEIR PROPERTIES

**DEFINITION** **Linear Transformation**

A map $T : \mathbb{R}^n \to \mathbb{R}^m$ is a **linear transformation** if for all vectors $\vec{u}, \vec{v} \in \mathbb{R}^n$ and all scalars $\lambda, \mu \in \mathbb{R}$:

$$T(\lambda\vec{u} + \mu\vec{v}) = \lambda T(\vec{u}) + \mu T(\vec{v}).$$

Equivalently, $T$ satisfies:

1. Additivity: $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$,
2. Homogeneity: $T(\lambda\vec{u}) = \lambda T(\vec{u})$.

**KEY POINTS** **Immediate consequences**

If $T$ is linear then:

- $T(\vec{0}) = \vec{0}$ (take $\lambda = \mu = 0$),
- $T(-\vec{u}) = -T(\vec{u})$,
- $T(\vec{u} - \vec{v}) = T(\vec{u}) - T(\vec{v})$.

**METHOD** **Proving that $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear**

1. Identify the formula. Write $T(\vec{x})$ explicitly in coordinates.
2. Quick test (necessary condition). Check whether $T(\vec{0}) = \vec{0}$. If not, $T$ is not linear: stop.
3. Prove additivity. Take arbitrary $\vec{u}, \vec{v}$ and show $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$.
4. Prove homogeneity. Take arbitrary $\lambda \in \mathbb{R}$ and show $T(\lambda\vec{u}) = \lambda T(\vec{u})$.
5. Conclude. Both conditions hold, so $T$ is linear. (Equivalently, verify in one step that $T(\lambda\vec{u} + \mu\vec{v}) = \lambda T(\vec{u}) + \mu T(\vec{v})$.)

Reflex: any map given by linear expressions in the coordinates (no constant term, no squares, no products of variables) is linear. Terms like $x^2$, $xy$, $x + 3$ are danger signals.

**EXEMPLE** **Testing linearity**

Determine which of the following maps are linear transformations. Justify your answers.

1. $T_1 : \mathbb{R}^2 \to \mathbb{R}^2, \quad T_1(x, y) = (2x - 3y,\ x + y)$.
2. $T_2 : \mathbb{R}^2 \to \mathbb{R}^2, \quad T_2(x, y) = (x + 1,\ y)$.
3. $T_3 : \mathbb{R}^2 \to \mathbb{R}, \quad T_3(x, y) = xy$.
4. $T_4 : \mathbb{R}^3 \to \mathbb{R}^2, \quad T_4(x, y, z) = (x + 2y - z,\ 3x - y + 4z)$.

**WORKED EXAMPLE 28** **Testing linearity**

Solution.

1. For $T_1$. First, $T_1(0, 0) = (0, 0)$, so the necessary condition passes. Let $\vec{u} = (x_1, y_1)$, $\vec{v} = (x_2, y_2)$ and $\lambda \in \mathbb{R}$.

$$\begin{aligned} T_1(\vec{u} + \vec{v}) &= T_1(x_1 + x_2,\ y_1 + y_2) \\ &= \big(2(x_1 + x_2) - 3(y_1 + y_2),\ (x_1 + x_2) + (y_1 + y_2)\big) \\ &= (2x_1 - 3y_1,\ x_1 + y_1) + (2x_2 - 3y_2,\ x_2 + y_2) = T_1(\vec{u}) + T_1(\vec{v}). \end{aligned}$$

   Also,

$$T_1(\lambda\vec{u}) = \big(2\lambda x_1 - 3\lambda y_1,\ \lambda x_1 + \lambda y_1\big) = \lambda T_1(\vec{u}).$$

   Both conditions hold, hence $T_1$ is a linear transformation.

2. For $T_2$. $T_2(0, 0) = (1, 0) \neq (0, 0)$. A linear transformation must send $\vec{0}$ to $\vec{0}$, so $T_2$ is not linear. (It is a translation, an affine map.)

3. For $T_3$. Although $T_3(0, 0) = 0$, take $\vec{u} = (1, 1)$ and $\lambda = 2$:

$$T_3(2\vec{u}) = T_3(2, 2) = 4, \qquad 2T_3(\vec{u}) = 2 \times 1 = 2.$$

   Since $T_3(2\vec{u}) \neq 2T_3(\vec{u})$, homogeneity fails and $T_3$ is not linear.

4. For $T_4$. $T_4(0, 0, 0) = (0, 0)$. Let $\vec{u} = (x_1, y_1, z_1)$, $\vec{v} = (x_2, y_2, z_2)$, $\lambda, \mu \in \mathbb{R}$.

$$\begin{aligned} T_4(\lambda\vec{u} + \mu\vec{v}) &= T_4(\lambda x_1 + \mu x_2,\ \lambda y_1 + \mu y_2,\ \lambda z_1 + \mu z_2) \\ &= \big((\lambda x_1 + \mu x_2) + 2(\lambda y_1 + \mu y_2) - (\lambda z_1 + \mu z_2), \\ &\qquad 3(\lambda x_1 + \mu x_2) - (\lambda y_1 + \mu y_2) + 4(\lambda z_1 + \mu z_2)\big) \\ &= \lambda(x_1 + 2y_1 - z_1,\ 3x_1 - y_1 + 4z_1) + \mu(x_2 + 2y_2 - z_2,\ 3x_2 - y_2 + 4z_2) \\ &= \lambda T_4(\vec{u}) + \mu T_4(\vec{v}). \end{aligned}$$

   Hence $T_4$ is a linear transformation.

**WARNING** **Common GCE traps**

- Forgetting to test $T(\vec{0}) = \vec{0}$ first — it can save time.
- Confusing linear with affine: $T(\vec{x}) = A\vec{x} + \vec{b}$ with $\vec{b} \neq \vec{0}$ is not linear.
- Assuming $T(\vec{u} \cdot \vec{v}) = T(\vec{u}) \cdot T(\vec{v})$ or $T(\vec{u} \times \vec{v}) = T(\vec{u}) \times T(\vec{v})$ — these are false in general.

## 13.1 Matrix of a Linear Transformation

**PROPERTY** **Matrix representation**

Every linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ can be represented by a unique $m \times n$ matrix $A$ such that $T(\vec{x}) = A\vec{x}$ for all $\vec{x} \in \mathbb{R}^n$. The columns of $A$ are the images of the standard basis vectors:

$$A = \begin{pmatrix} | & | & & | \\ T(\vec{e}_1) & T(\vec{e}_2) & \cdots & T(\vec{e}_n) \\ | & | & & | \end{pmatrix}.$$

**METHOD** **Finding the matrix of $T$ relative to the standard bases**

1. Compute $T(\vec{e}_1), T(\vec{e}_2), \ldots, T(\vec{e}_n)$ using the definition of $T$.
2. Write each image as a column vector.
3. Assemble the matrix $A$ by placing these columns side by side.
4. Check: verify on one arbitrary vector that $T(\vec{x}) = A\vec{x}$.

**WORKED EXAMPLE 29** **Matrix from the definition**

A linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$ is defined by

$$T(x, y, z) = (x + 2y - z,\ 3x + y + z).$$

Find the matrix $A$ of $T$ with respect to the standard bases, and compute $T(1, -1, 2)$.

**SOLUTION**

Solution. The standard basis of $\mathbb{R}^3$ is $\vec{e}_1 = (1, 0, 0)$, $\vec{e}_2 = (0, 1, 0)$, $\vec{e}_3 = (0, 0, 1)$. Compute the

images:

$$T(\vec{e}_1) = (1, 3), \qquad T(\vec{e}_2) = (2, 1), \qquad T(\vec{e}_3) = (-1, 1).$$

Placing these as columns:

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 3 & 1 & 1 \end{pmatrix}.$$

Check and computation:

$$T(1, -1, 2) = A \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1(1) + 2(-1) + (-1)(2) \\ 3(1) + 1(-1) + 1(2) \end{pmatrix} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}.$$

Direct verification: $T(1, -1, 2) = (1 - 2 - 2,\ 3 - 1 + 2) = (-3, 4)$. ✓

**EXEMPLE** **Matrix from images of a basis**

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation such that $T(1, 1) = (3, 2)$ and $T(1, -1) = (1, 4)$. Find the matrix of $T$ relative to the standard basis.

**WORKED EXAMPLE 30** **Matrix from images of a basis**

Solution. We need $T(1, 0)$ and $T(0, 1)$. Express the standard basis in terms of the given vectors:

$$(1, 0) = \tfrac{1}{2}(1, 1) + \tfrac{1}{2}(1, -1), \qquad (0, 1) = \tfrac{1}{2}(1, 1) - \tfrac{1}{2}(1, -1).$$

By linearity:

$$T(1, 0) = \tfrac{1}{2}T(1, 1) + \tfrac{1}{2}T(1, -1) = \tfrac{1}{2}(3, 2) + \tfrac{1}{2}(1, 4) = (2, 3),$$
$$T(0, 1) = \tfrac{1}{2}T(1, 1) - \tfrac{1}{2}T(1, -1) = \tfrac{1}{2}(3, 2) - \tfrac{1}{2}(1, 4) = (1, -1).$$

Hence the matrix is $A = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix}$.

**DID YOU KNOW?** **Application: Computer graphics**

Linear transformations are the backbone of 2D/3D computer graphics. Rotations, scaling, shearing, and reflections are all linear. In Cameroon's growing tech sector (e.g., animation

studios in Douala, game developers in Yaoundé), matrices like $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ rotate objects by angle $\theta$ — a direct application of the theory you are mastering.

## 14 KERNEL AND IMAGE OF A LINEAR TRANSFORMATION

**DEFINITION** **Kernel and Image**

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation with matrix $A$.

- The **kernel** of $T$ is $\ker T = \{\vec{x} \in \mathbb{R}^n : T(\vec{x}) = \vec{0}\} = \{\vec{x} : A\vec{x} = \vec{0}\}$.
- The **image** of $T$ is $\operatorname{Im} T = \{T(\vec{x}) : \vec{x} \in \mathbb{R}^n\} = \{A\vec{x} : \vec{x} \in \mathbb{R}^n\}$, the column space of $A$.

**PROPERTY** **Subspace properties**

$\ker T$ is a subspace of $\mathbb{R}^n$ (the domain). $\operatorname{Im} T$ is a subspace of $\mathbb{R}^m$ (the codomain).

**METHOD** **Finding $\ker T$ and $\operatorname{Im} T$**

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ have matrix $A$.

1. Kernel. Solve $A\vec{x} = \vec{0}$ (a homogeneous system) by row reduction (Gaussian elimination). Express the solution set with free parameters. This set is $\ker T$. Its dimension is the **nullity** $\nu(T)$.
2. Image. $\operatorname{Im} T$ is spanned by the columns of $A$: $\operatorname{Im} T = \operatorname{span}\{T(\vec{e}_1), \dots, T(\vec{e}_n)\}$. Extract a basis by keeping only linearly independent columns (pivot columns in the row-echelon form of $A$). Its dimension is the **rank** $\rho(T)$.
3. Verify the Rank–Nullity Theorem: $\dim \ker T + \dim \operatorname{Im} T = n$.
4. Conclusions: $T$ is injective (one-to-one) $\iff \ker T = \{\vec{0}\}$; $T$ is surjective (onto) $\iff \operatorname{Im} T = \mathbb{R}^m$.

**WORKED EXAMPLE 31** **Kernel and image**

Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be the linear transformation with matrix

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{pmatrix}.$$

Find a basis and the dimension of $\ker T$ and of $\operatorname{Im} T$. Is $T$ injective? Surjective?

SOLUTION

Solution.
Step 1: kernel. Solve $A\vec{x} = \vec{0}$. Row-reduce:

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{pmatrix} \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_3 \to R_3 - R_2} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

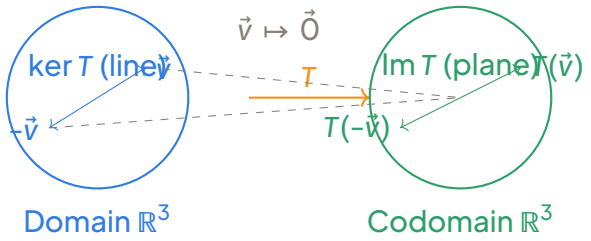
The system gives $z = 0$ and $x + 2y = 0$, i.e. $x = -2y$ with $y$ free. Setting $y = t$:

$$\ker T = \left\{ t \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} : t \in \mathbb{R} \right\} = \operatorname{span} \left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} \right\}, \qquad \dim \ker T = 1.$$

Step 2: image. The columns are $\vec{c}_1 = (1, 2, 1)$, $\vec{c}_2 = (2, 4, 2) = 2\vec{c}_1$, $\vec{c}_3 = (1, 3, 2)$. Since $\vec{c}_2$ is a multiple of $\vec{c}_1$ while $\vec{c}_1$ and $\vec{c}_3$ are not proportional, a basis is $\{\vec{c}_1, \vec{c}_3\}$:

$$\operatorname{Im} T = \operatorname{span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} \right\}, \qquad \dim \operatorname{Im} T = 2.$$

Step 3: verification. $\dim \ker T + \dim \operatorname{Im} T = 1 + 2 = 3 = \dim \mathbb{R}^3$. ✓
Conclusion. Since $\ker T \neq \{\vec{0}\}$, $T$ is not injective; since $\operatorname{Im} T \neq \mathbb{R}^3$ (dimension $2 < 3$), $T$ is not surjective. Geometrically, $T$ collapses $\mathbb{R}^3$ onto a plane.




Geometric interpretation: kernel vectors collapse to $\vec{0}$; the image is a plane.

**EXERCISE 8** **Practice: kernel and image**

Let $T:\mathbb{R}^4\to\mathbb{R}^3$ have matrix $A=\begin{pmatrix}1&2&0&1\\0&1&1&1\\1&3&1&2\end{pmatrix}$.

1. Find a basis for $\ker T$ and state its dimension.
2. Find a basis for $\operatorname{Im} T$ and state its dimension.
3. Verify the rank–nullity theorem.
4. Is $T$ injective? Surjective?

**SOLUTION** **Exercise 1**

Solution. Row-reduce $A$:

$$\begin{pmatrix}1&2&0&1\\0&1&1&1\\1&3&1&2\end{pmatrix}\xrightarrow{R_3\to R_3-R_1}\begin{pmatrix}1&2&0&1\\0&1&1&1\\0&1&1&1\end{pmatrix}\xrightarrow{R_3\to R_3-R_2}\begin{pmatrix}1&2&0&1\\0&1&1&1\\0&0&0&0\end{pmatrix}\xrightarrow{R_1\to R_1-2R_2}\begin{pmatrix}1&0&-2&-1\\0&1&1&1\\0&0&0&0\end{pmatrix}.$$

Pivot columns: 1 and 2. Free variables: $x_3=s$, $x_4=t$. System: $x_1-2x_3-x_4=0\Rightarrow x_1=2s+t$; $x_2+x_3+x_4=0\Rightarrow x_2=-s-t$.

$$\ker T=\left\{s\begin{pmatrix}2\\-1\\1\\0\end{pmatrix}+t\begin{pmatrix}1\\-1\\0\\1\end{pmatrix}:s,t\in\mathbb{R}\right\},\quad \dim\ker T=2.$$

Basis of $\operatorname{Im} T$: pivot columns of original $A$: $\begin{pmatrix}1\\0\\1\end{pmatrix},\begin{pmatrix}2\\1\\3\end{pmatrix}$. $\dim\operatorname{Im} T=2$. Rank–nullity: $2+2=4=\dim\mathbb{R}^4$. ✓Not injective ($\ker T\neq\{\vec{0}\}$), not surjective ($\dim\operatorname{Im} T=2<3$).

## 15 INVERSE OF A LINEAR TRANSFORMATION AND INVARIANT POINTS

PROPERTY **Invertibility criteria**

For a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^n$ with matrix $A$, the following are equivalent:

1. $T$ is invertible (bijective).
2. $A$ is invertible.
3. $\det A \neq 0$.
4. $\ker T = \{\vec{0}\}$.
5. $\operatorname{Im} T = \mathbb{R}^n$.
6. 0 is not an eigenvalue of $A$.

METHOD **Inverse transformation and invariant points**

Let $T$ have matrix $A$.

1. Invertibility. Check $\det A \neq 0$ (or $\ker T = \{\vec{0}\}$).
2. Inverse. Then $T^{-1}$ is linear with matrix $A^{-1}$. For a $2 \times 2$ matrix, $A^{-1} = \frac{1}{\det A}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.
3. Invariant (fixed) points. Solve $T(\vec{x}) = \vec{x}$, i.e. $(A - I)\vec{x} = \vec{0}$. The fixed points form a subspace (at least $\vec{0}$).
4. Interpretation. A line of fixed points means $T$ acts as the identity on that line (e.g. a reflection fixes its axis).

WARNING **Trap**

Do not confuse invariant points ($T(\vec{x}) = \vec{x}$, pointwise fixed) with invariant lines ($T$ maps the line into itself, points may move along it). Read the question carefully!

WORKED EXAMPLE 32 **Inverse and fixed points**

The linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ has matrix $A = \begin{pmatrix} 3 & -1 \\ 5 & -2 \end{pmatrix}$.

1. Show that $T$ is invertible and find the matrix of $T^{-1}$.
2. Determine all invariant points of $T$.

SOLUTION

Solution.

1. $\det A = (3)(-2) - (-1)(5) = -6 + 5 = -1 \neq 0$, so $A$ is invertible and $T$ is invertible.

$$A^{-1} = \frac{1}{-1}\begin{pmatrix} -2 & 1 \\ -5 & 3 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 5 & -3 \end{pmatrix}.$$

Hence $T^{-1}(x, y) = (2x - y,\ 5x - 3y)$. Check: $A^{-1}A = \begin{pmatrix} 2 & -1 \\ 5 & -3 \end{pmatrix}\begin{pmatrix} 3 & -1 \\ 5 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. ✓

2. Invariant points satisfy $(A - I)\vec{x} = \vec{0}$:

$$A - I = \begin{pmatrix} 2 & -1 \\ 5 & -3 \end{pmatrix}, \qquad \begin{cases} 2x - y = 0 \\ 5x - 3y = 0. \end{cases}$$

From the first equation $y = 2x$; substituting: $5x - 6x = -x = 0$, so $x = 0$, $y = 0$. The only invariant point is the origin $(0, 0)$.

EXEMPLE **Invariant points of a projection**

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the orthogonal projection onto the line $y = x$. Its matrix is $A = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$.

Find the invariant points.

WORKED EXAMPLE 33 **Invariant points of a projection**

Solution. Solve $(A - I)\vec{x} = \vec{0}$:

$$A - I = \frac{1}{2}\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \sim \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} \Rightarrow x - y = 0.$$

The invariant points are exactly the line $y = x$ (the axis of projection). Every point on this line is fixed; points off the line move onto it.

## 16 IMAGES OF LINES AND PLANES

METHOD **Image of a line or plane under $T$**

1. Parametrise the object: a line through $\vec{a}$ with direction $\vec{d}$ is $\vec{r} = \vec{a} + t\vec{d}$; a plane is $\vec{r} = \vec{a} + s\vec{u} + t\vec{v}$.
2. Apply linearity: $T(\vec{r}) = T(\vec{a}) + t\,T(\vec{d})$ (respectively $T(\vec{a}) + s\,T(\vec{u}) + t\,T(\vec{v})$).
3. Read off the image: the image is a line through $T(\vec{a})$ with direction $T(\vec{d})$ — unless $T(\vec{d}) = \vec{0}$, in which case the image is the single point $T(\vec{a})$.
4. For a plane: if $T(\vec{u})$ and $T(\vec{v})$ are independent, the image is a plane; if they are parallel but non-zero, a line; if both zero, a point.

Reflex: linear transformations map lines to lines (or points) and planes to planes (or lines or points): they can only lower the dimension, never raise it.

WORKED EXAMPLE 34 **Image of a line and of a plane**

Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be defined by $T(x, y, z) = (x + y,\ y + z,\ x + 2y + z)$.

1. Find the image of the line $L : \vec{r} = (1, 0, 2) + t(1, -1, 1)$.
2. Find the image of the plane $P : \vec{r} = (0, 0, 0) + s(1, 0, 0) + t(0, 1, 0)$.

SOLUTION

Solution. Note first that the matrix of $T$ is $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix}$, and the third row is the sum of the first two, so $\operatorname{rank} A = 2$: images will live in a plane.

1. Compute $T(1, 0, 2) = (1 + 0,\ 0 + 2,\ 1 + 0 + 2) = (1, 2, 3)$ and $T(1, -1, 1) = (1 - 1,\ -1 + 1,\ 1 - 2 + 1) = (0, 0, 0)$. The direction vector lies in $\ker T$! Hence

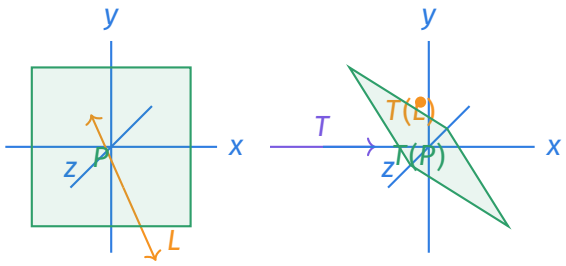
$$T(\vec{r}) = (1, 2, 3) + t(0, 0, 0) = (1, 2, 3).$$

   The image of $L$ is the single point $(1, 2, 3)$: the whole line collapses.

2. $T(1, 0, 0) = (1, 0, 1)$ and $T(0, 1, 0) = (1, 1, 2)$. These are not proportional, so they are independent. The image is the plane

$$\vec{r} = s(1, 0, 1) + t(1, 1, 2),$$

   i.e. the plane through the origin spanned by $(1, 0, 1)$ and $(1, 1, 2)$. (Cartesian form: a normal is $(1, 0, 1) \times (1, 1, 2) = (-1, -1, 1)$, giving $x + y - z = 0$.)




Line $L$ collapses to a point because its direction is in $\ker T$; plane $P$ maps to another plane.

**EXERCISE 9** **Practice: images of lines and planes**

Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ have matrix $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$.

1. Find the image of the line $\vec{r} = (1, 2, 3) + t(1, 1, -2)$.
2. Find the image of the plane $x + 2y - z = 4$.
3. Describe geometrically what $T$ does to $\mathbb{R}^3$.

**SOLUTION** **Exercise 2**

Solution. $T(x, y, z) = (x + z,\ y + z,\ 0)$.

1. $T(1, 2, 3) = (4, 5, 0)$, $T(1, 1, -2) = (1 - 2,\ 1 - 2,\ 0) = (-1, -1, 0) \neq \vec{0}$. Image: line $\vec{r} = (4, 5, 0) + t(-1, -1, 0)$ in the $xy$-plane.
2. Plane: $\vec{r} = (4, 0, 0) + s(1, 0, 1) + t(0, 1, 2)$ (since $z = x + 2y - 4$). $T(4, 0, 0) = (4, 0, 0)$, $T(1, 0, 1) = (2, 1, 0)$, $T(0, 1, 2) = (2, 3, 0)$. These two direction vectors are independent, so image is the whole $xy$-plane ($z = 0$).
3. $T$ projects $\mathbb{R}^3$ onto the $xy$-plane along the direction $(1, 1, -1)$ (kernel).

# 17 EIGENVALUES AND EIGENVECTORS

**DEFINITION** **Eigenvalues and Eigenvectors**

Let $A$ be an $n \times n$ matrix. A scalar $\lambda$ is an **eigenvalue** of $A$ if there exists a non-zero vector $\vec{v}$ such that $A\vec{v} = \lambda\vec{v}$. Such a vector $\vec{v}$ is called an **eigenvector** corresponding to $\lambda$. The set of all eigenvectors for $\lambda$ together with $\vec{0}$ is the **eigenspace** $E_\lambda = \ker(A - \lambda I)$.

PROPERTY **Characteristic equation**

The eigenvalues of $A$ are the roots of the **characteristic polynomial** $p(\lambda) = \det(A - \lambda I) = 0$.

METHOD **Finding eigenvalues and eigenvectors**

Given a square matrix $A$:

1. Characteristic equation. Compute $\det(A - \lambda I) = 0$ and solve for $\lambda$: the roots are the **eigenvalues**.
2. Eigenvectors. For each eigenvalue $\lambda_i$, solve $(A - \lambda_i I)\vec{x} = \vec{0}$. The non-zero solutions are the **eigenvectors**; the solution space is the **eigenspace** $E_{\lambda_i}$.
3. Checks. (i) Sum of eigenvalues $= \operatorname{tr}(A)$; (ii) product of eigenvalues $= \det A$; (iii) verify $A\vec{v} = \lambda\vec{v}$ on one eigenvector.
4. Geometric meaning. An eigenvector gives an invariant line through the origin: $T$ stretches it by the factor $\lambda$ (reversing direction if $\lambda < 0$).

WORKED EXAMPLE 35 **Eigenvalues and eigenvectors of a $2 \times 2$ matrix**

Find the eigenvalues and corresponding eigenvectors of $A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$, and describe the geometric action of the associated transformation on each eigenspace.

SOLUTION

Solution.
Step 1: eigenvalues.

$$\det(A - \lambda I) = \begin{vmatrix} 4 - \lambda & 2 \\ 1 & 3 - \lambda \end{vmatrix} = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = 0.$$

Factorising: $(\lambda - 2)(\lambda - 5) = 0$, so $\lambda_1 = 2, \lambda_2 = 5$. Check: $\operatorname{tr}(A) = 7 = 2 + 5$ ✓; $\det A = 12 - 2 = 10 = 2 \times 5$ ✓.

Step 2: eigenvectors for $\lambda_1 = 2$. Solve $(A - 2I)\vec{x} = \vec{0}$:

$$\begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \vec{0} \implies x + y = 0.$$

Eigenvectors: $\vec{v} = t\begin{pmatrix} 1 \\ -1 \end{pmatrix}$, $t \neq 0$. Eigenspace $E_2 = \text{span}\{(1, -1)\}$.

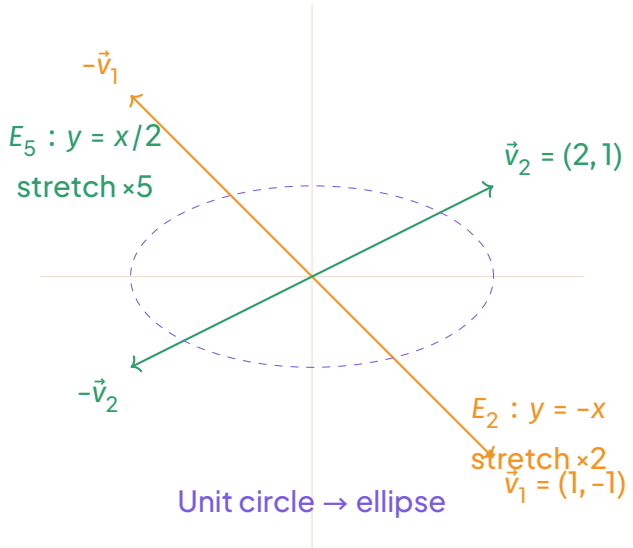
Step 3: eigenvectors for $\lambda_2 = 5$. Solve $(A - 5I)\vec{x} = \vec{0}$:

$$\begin{pmatrix} -1 & 2 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \vec{0} \implies x = 2y.$$

Eigenvectors: $\vec{v} = t\begin{pmatrix} 2 \\ 1 \end{pmatrix}$, $t \neq 0$. Eigenspace $E_5 = \text{span}\{(2, 1)\}$.

Step 4: verification. $A\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix} = 2\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ ✓; $A\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 10 \\ 5 \end{pmatrix} = 5\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ ✓.

Geometry. The line $y = -x$ is invariant, stretched by factor 2; the line $y = \frac{x}{2}$ is invariant, stretched by factor 5.




Eigenvectors define invariant lines; the transformation stretches by eigenvalues.

## 17.1 Diagonalisation of 3 × 3 Matrices

METHOD **3 × 3 case and diagonalisation**

1. Expand $\det(A - \lambda I)$ along a row or column (choose one with zeros). Look for an obvious root ($\lambda = 0, 1, 2, -1, \ldots$) and factorise.
2. Find a basis of each eigenspace by solving $(A - \lambda_i I)\vec{x} = \vec{0}$.
3. If three independent eigenvectors $\vec{v}_1, \vec{v}_2, \vec{v}_3$ exist, form $P = (\vec{v}_1\ \vec{v}_2\ \vec{v}_3)$; then $P^{-1}AP = D$ is diagonal with the eigenvalues on the diagonal, in the same order as the columns of $P$.
4. Application: $A^n = PD^nP^{-1}$, which makes powers easy to compute.

WARNING **Order matters**

The eigenvalues in $D$ must appear in the same order as the corresponding eigenvectors in $P$. Swapping columns of $P$ without swapping entries of $D$ is a classic GCE error.

WORKED EXAMPLE 36 **Diagonalising a 3 × 3 matrix**

Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$.

1. Find the eigenvalues of $A$.
2. Find a basis of each eigenspace.
3. Write down matrices $P$ and $D$ such that $A = PDP^{-1}$, and deduce $A^4$.

SOLUTION

Solution.

1. Expanding along the first row:

$$\det(A - \lambda I) = (1 - \lambda)\begin{vmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{vmatrix} = (1 - \lambda)\left[(2 - \lambda)^2 - 1\right] = (1 - \lambda)(\lambda^2 - 4\lambda + 3).$$

Since $\lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3)$, we get $\det(A - \lambda I) = (1 - \lambda)^2(3 - \lambda)\cdot(-1)^0$; precisely, the roots are $\lambda = 1$ (double) and $\lambda = 3$. Check: $\operatorname{tr}(A) = 5 = 1 + 1 + 3$ ✓.

2. For $\lambda = 1$: $(A - I)\vec{x} = \vec{0}$:

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}\vec{x} = \vec{0} \implies y + z = 0,\ x \text{ free.}$$

Two free parameters ($x$ and $y$, with $z = -y$): $E_1 = \text{span}\{(1, 0, 0), (0, 1, -1)\}$.

For $\lambda = 3$: $(A - 3I)\vec{x} = \vec{0}$:

$$\begin{pmatrix} -2 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{pmatrix}\vec{x} = \vec{0} \implies x = 0,\ y = z.$$

$E_3 = \text{span}\{(0, 1, 1)\}$.

3. We have three independent eigenvectors, so $A$ is diagonalisable:

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{pmatrix}, \qquad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Since the first two columns of $P$ lie in $E_1$ (eigenvalue 1), $A^4 = PD^4P^{-1}$ with $D^4 = \text{diag}(1, 1, 81)$.

The block structure gives directly: the $2 \times 2$ block $B = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ has eigenvalues 1, 3, and one computes $B^4 = \begin{pmatrix} 41 & 40 \\ 40 & 41 \end{pmatrix}$ (using $B^n = \frac{1}{2}\begin{pmatrix} 3^n + 1 & 3^n - 1 \\ 3^n - 1 & 3^n + 1 \end{pmatrix}$ with $3^4 = 81$). Hence

$$A^4 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 41 & 40 \\ 0 & 40 & 41 \end{pmatrix}.$$

DID YOU KNOW? **Diagonalisation in practice**

Diagonalisation is used to solve systems of linear differential equations, analyse stability in dynamical systems, and compute matrix powers efficiently. In structural engineering (e.g., designing bridges in Cameroon), the eigenvalues of stiffness matrices determine natural vibration frequencies — avoiding resonance is a matter of safety!

# 18 INTEGRATED GCE-STYLE PROBLEMS

**WORKED EXAMPLE 37** **Comprehensive problem: kernel, image, eigenvalues**

The linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ has matrix

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

1. Show that $\vec{v}_1 = (1, 1, 1)$ is an eigenvector and find its eigenvalue.
2. Show that every vector of the plane $x + y + z = 0$ is an eigenvector with eigenvalue 1.
3. Deduce the eigenvalues of $A$ and a basis of eigenvectors.
4. Find $\ker(T - I)$ and state the set of invariant points of $T$.
5. Is $T$ invertible? Justify.

**SOLUTION**

Solution.

1. $A\vec{v}_1 = \begin{pmatrix} 2+1+1 \\ 1+2+1 \\ 1+1+2 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \\ 4 \end{pmatrix} = 4\vec{v}_1$. So $\vec{v}_1$ is an eigenvector with eigenvalue $\lambda = 4$.

2. Let $\vec{x} = (x, y, z)$ with $x + y + z = 0$. Then

$$A\vec{x} = \begin{pmatrix} 2x + y + z \\ x + 2y + z \\ x + y + 2z \end{pmatrix} = \begin{pmatrix} x + (x + y + z) \\ y + (x + y + z) \\ z + (x + y + z) \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \vec{x}.$$

   Every vector of the plane $x + y + z = 0$ is fixed by $A$, i.e. is an eigenvector with eigenvalue $\lambda = 1$.

3. The plane $x + y + z = 0$ has dimension 2 (basis e.g. $(1, -1, 0), (1, 0, -1)$) and $\vec{v}_1$ is not in it, so $\{(1, 1, 1), (1, -1, 0), (1, 0, -1)\}$ is a basis of $\mathbb{R}^3$ of eigenvectors. Eigenvalues: 1 (double) and 4. Check: $\operatorname{tr}(A) = 6 = 1 + 1 + 4$ ✓.

4. $\ker(T - I) = E_1 = \{(x, y, z) : x + y + z = 0\}$, the plane through the origin with normal $(1, 1, 1)$. Since $(T - I)\vec{x} = \vec{0} \iff T(\vec{x}) = \vec{x}$, the set of invariant points of $T$ is exactly this plane.

5. $\det A$ = product of eigenvalues $= 1 \times 1 \times 4 = 4 \neq 0$, so $A$ is invertible and $T$ is invertible. (Equivalently, $\ker T = \{\vec{0}\}$ since 0 is not an eigenvalue.)

**WORKED EXAMPLE 38** **GCE 2022 style: Invariant lines and diagonalisation**

A linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ is represented by the matrix

$$M = \begin{pmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{pmatrix}.$$

1. Find the eigenvalues of $M$.
2. For each eigenvalue, find a basis of the corresponding eigenspace.
3. Write down a matrix $P$ and a diagonal matrix $D$ such that $M = PDP^{-1}$.
4. Hence find $M^n$ for any positive integer $n$.
5. Determine the Cartesian equations of all invariant lines of $T$ through the origin.

**SOLUTION**

Solution.

1. $\det(M - \lambda I) = \begin{vmatrix} 4-\lambda & -2 & 2 \\ -2 & 1-\lambda & -1 \\ 2 & -1 & 1-\lambda \end{vmatrix}$. Expand along first row:

$$\begin{aligned} &(4-\lambda)\left[(1-\lambda)^2 - 1\right] + 2\left[-2(1-\lambda) + 2\right] + 2\left[2 - 2(1-\lambda)\right] \\ &= (4-\lambda)(\lambda^2 - 2\lambda) + 2(2\lambda) + 2(2\lambda) \\ &= (4-\lambda)\lambda(\lambda - 2) + 8\lambda = \lambda\left[(4-\lambda)(\lambda-2) + 8\right] \\ &= \lambda\left[-\lambda^2 + 6\lambda - 8 + 8\right] = \lambda(-\lambda^2 + 6\lambda) = -\lambda^2(\lambda - 6). \end{aligned}$$

Eigenvalues: $\lambda = 0$ (double), $\lambda = 6$.

2. For $\lambda = 0$: Solve $M\vec{x} = \vec{0}$:

$$\begin{pmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{pmatrix} \sim \begin{pmatrix} 2 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \implies 2x - y + z = 0.$$

Two free parameters: $E_0 = \text{span}\{(1, 2, 0), (1, 0, -2)\}$.

For $\lambda = 6$: Solve $(M - 6I)\vec{x} = \vec{0}$:

$$\begin{pmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & -1 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{pmatrix} \implies x + y - z = 0,\ y + z = 0.$$

Solution: $y = -z$, $x = 2z$. $E_6 = \text{span}\{(2, -1, 1)\}$.

3. $P = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 0 & -1 \\ 0 & -2 & 1 \end{pmatrix}$, $D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 6 \end{pmatrix}$ (order matches columns of $P$).

4. $M^n = PD^nP^{-1}$ with $D^n = \operatorname{diag}(0, 0, 6^n)$. Compute $P^{-1}$:

$$P^{-1} = \frac{1}{9}\begin{pmatrix} -2 & 5 & -1 \\ -2 & -1 & 5 \\ 4 & 2 & 2 \end{pmatrix}.$$

Then $M^n = 6^n \cdot \frac{1}{9}\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}\begin{pmatrix} 4 & 2 & 2 \end{pmatrix} = \frac{6^n}{9}\begin{pmatrix} 8 & 4 & 4 \\ -4 & -2 & -2 \\ 4 & 2 & 2 \end{pmatrix} = \frac{2 \cdot 6^{n-1}}{3}\begin{pmatrix} 4 & 2 & 2 \\ -2 & -1 & -1 \\ 2 & 1 & 1 \end{pmatrix}$.

5. Invariant lines through origin = eigenspaces. $\lambda = 0$: plane $2x - y + z = 0$ (every line in this plane is invariant, mapped to $\vec{0}$). $\lambda = 6$: line $\frac{x}{2} = \frac{y}{-1} = \frac{z}{1}$ (stretched by factor 6).

**EXERCISE 10** **Final practice: mixed topics**

Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be defined by $T(x, y, z) = (2x + y,\ x + 2y,\ 3z)$.

1. Find the matrix $A$ of $T$.
2. Find $\ker T$ and $\operatorname{Im} T$. State their dimensions.
3. Find all eigenvalues and a basis for each eigenspace.
4. Is $T$ diagonalisable? If yes, find $P$ and $D$ such that $A = PDP^{-1}$.
5. Find the image of the line $\vec{r} = (1, 1, 1) + t(1, -1, 0)$.
6. Find the invariant points of $T$.

**SOLUTION** **Exercise 3**

Solution.

1. $A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$.

2. $A\vec{x} = \vec{0} \Rightarrow 2x + y = 0,\ x + 2y = 0,\ 3z = 0 \Rightarrow x = y = z = 0$. $\ker T = \{\vec{0}\}$, dim = 0. $\operatorname{Im} T = \mathbb{R}^3$,

dim = 3. **Rank–nullity:** 0 + 3 = 3.

3. $\det(A - \lambda I) = (3 - \lambda)\begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = (3 - \lambda)(\lambda^2 - 4\lambda + 3) = (3 - \lambda)^2(1 - \lambda)$. **Eigenvalues:** $\lambda = 3$ **(double)**, $\lambda = 1$. $\lambda = 3$: $(A - 3I)\vec{x} = \vec{0} \Rightarrow -x + y = 0,\ x - y = 0,\ 0 = 0 \Rightarrow y = x,\ z$ **free.** $E_3 = \text{span}\{(1, 1, 0), (0, 0, 1)\}$. $\lambda = 1$: $(A - I)\vec{x} = \vec{0} \Rightarrow x + y = 0,\ x + y = 0,\ 2z = 0 \Rightarrow y = -x,\ z = 0$. $E_1 = \text{span}\{(1, -1, 0)\}$.

4. **Yes, three independent eigenvectors.** $P = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

5. $T(1, 1, 1) = (3, 3, 3)$, $T(1, -1, 0) = (1, -1, 0)$. **Image: line** $\vec{r} = (3, 3, 3) + t(1, -1, 0)$.

6. **Invariant points:** $(A - I)\vec{x} = \vec{0} \Rightarrow x + y = 0,\ 2z = 0 \Rightarrow y = -x,\ z = 0$. **Line** $\frac{x}{1} = \frac{y}{-1},\ z = 0$ **(the eigenspace** $E_1$**).**

**KEY POINTS** **Chapter summary: Linear Transformations**

- **Linearity:** $T(\lambda\vec{u} + \mu\vec{v}) = \lambda T(\vec{u}) + \mu T(\vec{v})$. **Test** $T(\vec{0}) = \vec{0}$ **first.**
- **Matrix: Columns are** $T(\vec{e}_i)$. $T(\vec{x}) = A\vec{x}$.
- **Kernel: Solve** $A\vec{x} = \vec{0}$. $\dim \ker T$ **= nullity.**
- **Image: Column space of** $A$. $\dim \text{Im}\, T$ **= rank.**
- **Rank–Nullity:** $\rho(T) + \nu(T) = n$.
- **Inverse: Exists** $\iff \det A \neq 0 \iff \ker T = \{\vec{0}\}$. **Matrix** $A^{-1}$.
- **Invariant points: Solve** $(A - I)\vec{x} = \vec{0}$.
- **Images of lines/planes: Apply** $T$ **to parametric form. Dimension never increases.**
- **Eigenvalues: Roots of** $\det(A - \lambda I) = 0$.
- **Eigenvectors: Non-zero solutions of** $(A - \lambda I)\vec{x} = \vec{0}$.
- **Diagonalisation:** $A = PDP^{-1}$ **if** $n$ **independent eigenvectors exist.** $A^n = PD^nP^{-1}$.

# 19 EXERCISES

## 19.1 I check my knowledge

**EXERCISE 11** **Multiple choice questions**

For each question, choose the single correct answer, justifying briefly.

1. Which of the following mappings $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation?

   (a) $T(x, y) = (2x - y,\ x + 3y)$

   (b) $T(x, y) = (x^2,\ y)$

   (c) $T(x, y) = (x + y,\ 1)$

   (d) $T(x, y) = (xy,\ x - y)$

2. The kernel of the linear transformation $T(x, y) = (x - y,\ 2x - 2y)$ is:

   (a) $\{(0, 0)\}$

   (b) the line $y = x$

   (c) the line $y = 2x$

   (d) the whole plane $\mathbb{R}^2$

3. A linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is bijective if and only if:

   (a) $\ker T = \mathbb{R}^2$

   (b) $\operatorname{Im} T = \{(0, 0)\}$

   (c) $\ker T = \{(0, 0)\}$

   (d) $\det T = 0$

4. The eigenvalues of $A = \begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix}$ are:

   (a) 3 and -5

   (b) 3 and 5

   (c) 8 and 0

   (d) 15 and 0

**EXERCISE 12** **True or false**

State whether each assertion is true or false, justifying your answer.

1. Every linear transformation maps the zero vector to the zero vector.
2. If $T(u + v) = T(u) + T(v)$ for all vectors $u, v$, then $T$ is necessarily linear.
3. The kernel of a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is always a line through the origin.
4. If $T$ is a linear transformation of the plane with $\det T \neq 0$, then the image of any line is a line.
5. An eigenvector of a linear transformation can never be the zero vector.

**EXERCISE 13** **Key definitions**

1. Define a **linear transformation** from $\mathbb{R}^2$ to $\mathbb{R}^2$.
2. Define the **kernel** and the **image** of a linear transformation $T$.
3. Define an **eigenvalue** and an **eigenvector** of a linear transformation.
4. What is an **invariant point** (fixed point) of a transformation?

**EXERCISE 14** **Direct calculations**

1. Given $T(x, y) = (3x + y,\ x - 2y)$, compute $T(1, 2)$, $T(-1, 3)$ and $T(0, 0)$.
2. Write down the matrix of $T(x, y) = (3x + y,\ x - 2y)$ and compute its determinant.
3. A linear transformation of the plane maps $(1, 0)$ to $(3, -1)$ and $(0, 1)$ to $(2, 4)$. Write down its matrix and find the image of $(5, 2)$.
4. Compute the characteristic polynomial of $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

## 19.2 I practise

**EXERCISE 15** **Testing linearity**

Determine whether each of the following mappings is a linear transformation, justifying

your answer carefully:

1. $T(x, y) = (2x - y,\ x + 3y)$;
2. $S(x, y) = (x^2,\ y)$;
3. $U(x, y) = (x + y,\ 1)$;
4. $V(x, y) = (x - 2y,\ 0,\ 3x + y)$ from $\mathbb{R}^2$ to $\mathbb{R}^3$.

**EXERCISE 16** **Kernel and image**

Let $T(x, y) = (x + 2y,\ 2x + 4y)$.

1. Find $\ker T$ and describe it geometrically.
2. Find $\operatorname{Im} T$ and describe it geometrically.
3. State, with reasons, whether $T$ is injective and whether $T$ is surjective.
4. Verify that $\dim(\ker T) + \dim(\operatorname{Im} T) = 2$.

**EXERCISE 17** **Inverse and invariant points**

Consider the transformation with matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

1. Show that the transformation is bijective.
2. Find its inverse transformation, i.e. express $T^{-1}(x', y')$ in terms of $x'$ and $y'$.
3. Determine all the invariant points of $T$.

**EXERCISE 18** **Image of a line**

Find the image of the line $2x - y + 3 = 0$ under the transformation $T(x, y) = (x - y,\ 2x + y)$, giving a Cartesian equation of the image line. Verify your answer by checking that the images of two points of the original line satisfy your equation.

## 19.3 I prepare for the GCE

**EXERCISE 19** **GCE style: a shear transformation**

(12 marks)

$T$ is the linear transformation of the plane with matrix $M = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$.

1. Identify $T$ geometrically, stating precisely the direction and the factor of the transformation. (3 marks)
2. Find the images of the vertices of the unit square with vertices $(0, 0)$, $(1, 0)$, $(1, 1)$, $(0, 1)$, sketch the image figure, and compute its area. (4 marks)
3. Find all the fixed points of $T$. (2 marks)
4. Find all the eigenvalues of $M$ and the corresponding eigenvectors. Comment on the link with part (c). (3 marks)

**EXERCISE 20** **GCE style: transformation of space**

(14 marks)

In space, the linear transformation $T$ is defined by

$$T(x, y, z) = (x + y,\ y + z,\ x + 2y + z).$$

1. Write down the matrix of $T$ and compute its determinant. (3 marks)
2. Find the kernel of $T$ and describe it geometrically. (3 marks)
3. Find the image of $T$ and show that it is a plane through the origin, giving a Cartesian equation of this plane. (4 marks)
4. Determine the image of the plane $z = 0$ under $T$. (2 marks)
5. Is $T$ bijective? Justify your answer in two different ways. (2 marks)

**EXERCISE 21** **GCE style: eigenvalues, powers and iteration**

(16 marks)

Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and let $T$ be the associated linear transformation of the plane.

1. Find the eigenvalues $\lambda_1, \lambda_2$ of $A$ and a corresponding eigenvector for each. (4 marks)
2. A transformation of the plane has eigenvalues $3$ and $-1$ with eigenvectors $(1, 1)$ and $(1, -1)$

respectively. By first expressing $(4, 0)$ as a linear combination of the eigenvectors, find the image of $(4, 0)$. (4 marks)

3. Returning to the matrix $A$ above, conjecture a formula for $A^n$ of the form

$$A^n = \frac{1}{2}\begin{pmatrix} 3^n + 1 & 3^n - 1 \\ 3^n - 1 & 3^n + 1 \end{pmatrix},$$

verify it for $n = 1$ and $n = 2$, and prove it by induction on $n$. (5 marks)

4. Hence compute the image of the vector $(1, 0)$ when $T$ is applied ten times successively. (3 marks)

# 20 SOLUTIONS TO THE EXERCISES

**SOLUTION** **Exercise 1 — Multiple choice questions**

1. Answer (a). $T(x, y) = (2x - y,\ x + 3y)$ has the form $(ax + by,\ cx + dy)$, hence it is linear (its matrix is $\begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$). The others fail: (b) contains $x^2$ (e.g. $T(2, 0) = (4, 0) \neq 2T(1, 0) = (2, 0)$); (c) gives $T(0, 0) = (0, 1) \neq (0, 0)$; (d) contains the product $xy$ (e.g. $T(2, 2) = (4, 0) \neq 2T(1, 1) = (2, 0)$).
2. Answer (b). $T(x, y) = (0, 0) \iff x - y = 0 \iff y = x$, a line through the origin.
3. Answer (c). For a linear transformation between spaces of the same finite dimension, bijectivity is equivalent to injectivity, which is equivalent to $\ker T = \{(0, 0)\}$ (equivalently $\det T \neq 0$, not 0).
4. Answer (b). A diagonal matrix has its diagonal entries as eigenvalues: $\lambda = 3$ and $\lambda = 5$ (the characteristic polynomial is $(3 - \lambda)(5 - \lambda)$).

**SOLUTION** **Exercise 2 — True or false**

1. True. $T(0) = T(0 + 0) = T(0) + T(0)$, and subtracting $T(0)$ from both sides gives $T(0) = 0$.
2. False. Additivity alone does not imply homogeneity $T(\lambda u) = \lambda T(u)$. For example, on $\mathbb{R}$ viewed as a vector space over $\mathbb{Q}$, additive non-linear maps exist; more simply at this level, one must also check $T(\lambda u) = \lambda T(u)$ — the two conditions together define linearity.
3. False. The kernel is always a vector subspace, but it can be $\{(0, 0)\}$ (e.g. the identity), a line through the origin, or the whole plane $\mathbb{R}^2$ (the zero transformation). It is not always a line.
4. True. If $\det T \neq 0$, $T$ is bijective and maps lines to lines: the image of a line with direction vector $u$ passing through $A$ is the line through $T(A)$ with direction vector $T(u) \neq 0$.
5. True. By definition an eigenvector is a non-zero vector $u$ such that $T(u) = \lambda u$. The zero vector is excluded (otherwise every scalar would be an eigenvalue).

**SOLUTION** **Exercise 3 — Key definitions**

1. A mapping $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a **linear transformation** if for all vectors $u, v \in \mathbb{R}^2$ and every

scalar $\lambda \in \mathbb{R}$:

$$T(u+v) = T(u) + T(v) \qquad \text{and} \qquad T(\lambda u) = \lambda\, T(u).$$

Equivalently, $T(\lambda u + \mu v) = \lambda T(u) + \mu T(v)$ for all $u, v$ and all scalars $\lambda, \mu$.

2. The **kernel** of $T$ is $\ker T = \{\, u \in \mathbb{R}^2 \;:\; T(u) = 0 \,\}$, the set of vectors mapped to the zero vector. The **image** of $T$ is $\operatorname{Im} T = \{\, T(u) \;:\; u \in \mathbb{R}^2 \,\}$, the set of all values taken by $T$. Both are vector subspaces.

3. A scalar $\lambda$ is an **eigenvalue** of $T$ if there exists a non-zero vector $u$ such that $T(u) = \lambda u$; such a vector $u$ is then called an **eigenvector** associated with $\lambda$.

4. An **invariant point** (fixed point) of a transformation $T$ is a point $M$ such that $T(M) = M$: the point is mapped onto itself.

**SOLUTION** **Exercise 4 — Direct calculations**

1. $T(1,2) = (3(1)+2,\ 1-2(2)) = (5,-3)$; $\quad T(-1,3) = (3(-1)+3,\ -1-2(3)) = (0,-7)$; $\quad T(0,0) = (0,0)$, as expected for a linear transformation.

2. The matrix is read from the columns $T(1,0) = (3,1)$ and $T(0,1) = (1,-2)$:

$$A = \begin{pmatrix} 3 & 1 \\ 1 & -2 \end{pmatrix}, \qquad \det A = (3)(-2) - (1)(1) = -7.$$

3. The columns of the matrix are the images of the basis vectors:

$$A = \begin{pmatrix} 3 & 2 \\ -1 & 4 \end{pmatrix}, \qquad T(5,2) = \begin{pmatrix} 3 & 2 \\ -1 & 4 \end{pmatrix}\begin{pmatrix} 5 \\ 2 \end{pmatrix} = \begin{pmatrix} 15+4 \\ -5+8 \end{pmatrix} = \begin{pmatrix} 19 \\ 3 \end{pmatrix}.$$

So $T(5,2) = (19,3)$.

4. The characteristic polynomial is

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3.$$

**SOLUTION** **Exercise 5 — Testing linearity**

1. $T$ is linear. Let $u = (x_1, y_1)$, $v = (x_2, y_2)$ and $\lambda \in \mathbb{R}$. Then

$$T(u+v) = (2(x_1+x_2) - (y_1+y_2),\ (x_1+x_2) + 3(y_1+y_2)) = T(u) + T(v),$$

and $T(\lambda u) = (2\lambda x_1 - \lambda y_1,\ \lambda x_1 + 3\lambda y_1) = \lambda T(u)$. Both axioms hold (equivalently, $T$ has the form $(ax + by, cx + dy)$).

2. $S$ is not linear. Counter-example to homogeneity: $S(2, 0) = (4, 0)$ but $2S(1, 0) = 2(1, 0) = (2, 0)$. Since $S(2u) \neq 2S(u)$ for $u = (1, 0)$, $S$ fails.

3. $U$ is not linear. $U(0, 0) = (0, 1) \neq (0, 0)$, whereas every linear transformation maps the zero vector to the zero vector.

4. $V$ is linear. Each component is a linear form in $x$ and $y$: $V(x, y) = (x - 2y,\ 0,\ 3x + y)$. Checking: $V(u + v) = V(u) + V(v)$ and $V(\lambda u) = \lambda V(u)$ follow from distributivity of scalar multiplication in each coordinate. Its matrix is $\begin{pmatrix} 1 & -2 \\ 0 & 0 \\ 3 & 1 \end{pmatrix}$.

**SOLUTION** **Exercise 6 — Kernel and image**

1. $(x, y) \in \ker T \iff \begin{cases} x + 2y = 0 \\ 2x + 4y = 0 \end{cases} \iff x = -2y$. Hence

$$\ker T = \{(-2t,\ t) \ :\ t \in \mathbb{R}\},$$

the line through the origin with direction vector $(-2, 1)$ (equation $x + 2y = 0$).

2. $T(x, y) = (x + 2y,\ 2(x + 2y))$: the second coordinate is always twice the first. Conversely, for any $s$, $(s, 2s) = T(s, 0)$. Hence

$$\operatorname{Im} T = \{(s, 2s) \ :\ s \in \mathbb{R}\},$$

the line through the origin of equation $y = 2x$, spanned by $(1, 2)$.

3. $T$ is not injective because $\ker T \neq \{(0, 0)\}$ (for instance $T(-2, 1) = (0, 0) = T(0, 0)$). $T$ is not surjective because $\operatorname{Im} T \neq \mathbb{R}^2$ (for instance $(1, 0)$ has no pre-image, since it would require $2x + 4y = 0$ and $x + 2y = 1$ simultaneously).

4. $\ker T$ is a line, so $\dim(\ker T) = 1$; $\operatorname{Im} T$ is a line, so $\dim(\operatorname{Im} T) = 1$. Then

$$\dim(\ker T) + \dim(\operatorname{Im} T) = 1 + 1 = 2 = \dim \mathbb{R}^2,$$

confirming the rank–nullity theorem.

**SOLUTION** **Exercise 7 — Inverse and invariant points**

1. $\det A = (2)(2) - (1)(1) = 3 \neq 0$, so $A$ is invertible and the transformation $T$ is bijective.

2. $A^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. Therefore, writing $(x', y') = T(x, y)$,

$$T^{-1}(x', y') = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} x' \\ y' \end{pmatrix} = \left(\frac{2x' - y'}{3},\ \frac{-x' + 2y'}{3}\right).$$

   Check: $T^{-1}(T(x, y)) = \frac{1}{3}\big(2(2x + y) - (x + 2y),\ -(2x + y) + 2(x + 2y)\big) = (x, y)$. ✓

3. Invariant points satisfy $T(x, y) = (x, y)$, i.e.

$$\begin{cases} 2x + y = x \\ x + 2y = y \end{cases} \iff \begin{cases} x + y = 0 \\ x + y = 0 \end{cases} \iff y = -x.$$

   The set of invariant points is the line $y = -x$ through the origin: every point of this line is fixed by $T$.

**SOLUTION** **Exercise 8 — Image of a line**

Let $(x', y') = T(x, y) = (x - y,\ 2x + y)$. Since $\det T = 1 \cdot 1 - (-1) \cdot 2 = 3 \neq 0$, $T$ is invertible and we can express $x, y$ in terms of $x', y'$:

$$\begin{cases} x' = x - y \\ y' = 2x + y \end{cases} \implies x = \frac{x' + y'}{3}, \qquad y = \frac{y' - 2x'}{3}.$$

Substituting into $2x - y + 3 = 0$:

$$2\left(\frac{x' + y'}{3}\right) - \left(\frac{y' - 2x'}{3}\right) + 3 = 0 \iff \frac{2x' + 2y' - y' + 2x'}{3} + 3 = 0 \iff 4x' + y' + 9 = 0.$$

Hence the image is the line $\boxed{4x + y + 9 = 0}$.
Verification. Choose two points of $2x - y + 3 = 0$: $A = (0, 3)$ and $B = (-1, 1)$.

$$T(A) = (0 - 3,\ 0 + 3) = (-3, 3): \quad 4(-3) + 3 + 9 = 0 \checkmark$$

$$T(B) = (-1 - 1,\ -2 + 1) = (-2, -1): \quad 4(-2) - 1 + 9 = 0 \checkmark$$

Both images satisfy $4x + y + 9 = 0$, confirming the result.

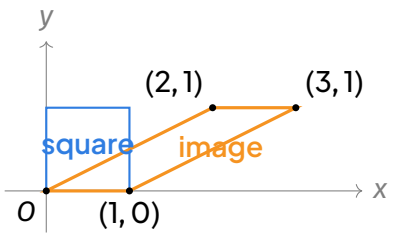
**SOLUTION** **Exercise 9 — GCE style: a shear transformation**

1. $T(x, y) = (x + 2y,\ y)$. This is a shear parallel to the $x$-axis (direction: the horizontal axis, along the vector $(1, 0)$), with factor (shear ratio) $k = 2$: each point is displaced horizontally by $2y$, proportional to its distance from the $x$-axis. Points of the $x$-axis are fixed. [3 marks: direction 1, factor 1, description 1]

2. Images of the vertices:

$$(0, 0) \mapsto (0, 0), \quad (1, 0) \mapsto (1, 0), \quad (1, 1) \mapsto (3, 1), \quad (0, 1) \mapsto (2, 1).$$

The image is a parallelogram with base 1 (from $(0, 0)$ to $(1, 0)$) and height 1, so its area is $1 \times 1 = 1$ square unit. (Equivalently, $|\det M| = 1$, so areas are preserved.)




The unit square (blue) and its image under the shear (orange).

[4 marks: images 2, sketch 1, area 1]

3. Fixed points: $T(x, y) = (x, y) \iff x + 2y = x \iff y = 0$. The set of fixed points is the $x$-axis (the line $y = 0$). [2 marks]

4. Characteristic polynomial: $\det(M - \lambda I) = (1 - \lambda)^2 = 0$, so the only eigenvalue is $\lambda = 1$ (double). Eigenvectors: $(M - I)\binom{x}{y} = 0 \iff 2y = 0 \iff y = 0$: the eigenvectors are the non-zero multiples of $(1, 0)$.

   Link with (c): eigenvectors with eigenvalue 1 are exactly the non-zero fixed directions; since every point of the $x$-axis is fixed, the $x$-axis is both the line of fixed points and the eigenspace for $\lambda = 1$. [3 marks: eigenvalue 1, eigenvectors 1, comment 1]

**SOLUTION** **Exercise 10 — GCE style: transformation of space**

1. Reading the columns $T(1, 0, 0) = (1, 0, 1)$, $T(0, 1, 0) = (1, 1, 2)$, $T(0, 0, 1) = (0, 1, 1)$:

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix}, \qquad \det A = 1\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} - 1\begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = 1(1 - 2) - 1(0 - 1) = -1 + 1 = 0.$$

[3 marks: matrix 2, determinant 1]

2. $(x,y,z) \in \ker T \iff \begin{cases} x+y=0 \\ y+z=0 \\ x+2y+z=0 \end{cases}$. The first two give $x=-y$, $z=-y$; the third is then $-y+2y-y=0$, automatically satisfied. Hence

$$\ker T = \{(-t,\ t,\ -t)\ :\ t \in \mathbb{R}\},$$

the line through the origin with direction vector $(-1,1,-1)$. [3 marks]

3. $T(x,y,z) = (x+y,\ y+z,\ x+2y+z)$. Note that the third coordinate equals the sum of the first two: $(x+y)+(y+z) = x+2y+z$. Hence every image vector $(X,Y,Z)$ satisfies $Z = X+Y$. Conversely, given $(X,Y,Z)$ with $Z=X+Y$, take $(x,y,z)=(X,0,Y)$: then $T(X,0,Y) = (X,Y,X+Y) = (X,Y,Z)$. Therefore

$$\operatorname{Im} T = \{(X,Y,Z)\ :\ X+Y-Z=0\},$$

the plane through the origin with Cartesian equation $x+y-z=0$ (normal vector $(1,1,-1)$). [4 marks]

4. Points of the plane $z=0$ are $(x,y,0)$; their images are $T(x,y,0) = (x+y,\ y,\ x+2y)$. As $(x,y)$ ranges over $\mathbb{R}^2$, the parameters $u = x+y$ and $v = y$ range over all of $\mathbb{R}^2$ (the map $(x,y) \mapsto (u,v)$ is invertible: $y=v$, $x=u-v$), and the third coordinate is $x+2y = u+v$. Hence the image is

$$\{(u,v,u+v)\ :\ u,v \in \mathbb{R}\},$$

i.e. the plane $z=x+y$, which is exactly $\operatorname{Im} T$ itself. [2 marks]

5. $T$ is not bijective.

   - First way: $\det A = 0$, so $A$ is not invertible, so $T$ is not bijective.
   - Second way: $\ker T \neq \{(0,0,0)\}$ (it is a whole line), so $T$ is not injective — for instance $T(-1,1,-1) = (0,0,0) = T(0,0,0)$ — hence not bijective. (Equivalently, $\operatorname{Im} T \neq \mathbb{R}^3$, so $T$ is not surjective.)

[2 marks: 1 per justification]

**SOLUTION** **Exercise 11 — GCE style: eigenvalues, powers and iteration**

1. Characteristic polynomial: $\det(A-\lambda I) = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda-1)(\lambda-3)$, so $\lambda_1 = 1$, $\lambda_2 = 3$.

   For $\lambda = 1$: $(A-I)\binom{x}{y} = 0 \iff x+y=0$, eigenvector $u_1 = (1,-1)$.

   For $\lambda = 3$: $(A-3I)\binom{x}{y} = 0 \iff -x+y=0$, eigenvector $u_2 = (1,1)$. [4 marks: polynomial 1,

eigenvalues 1, eigenvectors 2]

2. Write $(4,0) = \alpha(1,1) + \beta(1,-1)$: then $\alpha + \beta = 4$ and $\alpha - \beta = 0$, giving $\alpha = \beta = 2$. So $(4,0) = 2(1,1) + 2(1,-1)$. By linearity and the eigenvector property $T(u) = \lambda u$:

$$T(4,0) = 2\,T(1,1) + 2\,T(1,-1) = 2 \cdot 3(1,1) + 2 \cdot 1(1,-1) = (6,6) + (2,-2) = (8,4).$$

[4 marks: decomposition 2, use of eigenvalues 1, answer 1]

3. Verification. For $n = 1$: $\frac{1}{2}\begin{pmatrix} 3+1 & 3-1 \\ 3-1 & 3+1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = A$. ✓

For $n = 2$: $A^2 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$, and the formula gives $\frac{1}{2}\begin{pmatrix} 9+1 & 9-1 \\ 9-1 & 9+1 \end{pmatrix} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$. ✓

Induction. Let $P(n)$ be the stated formula. $P(1)$ holds (shown above). Assume $P(n)$ for some $n \geq 1$. Then

$$\begin{aligned} A^{n+1} = A^n A &= \frac{1}{2}\begin{pmatrix} 3^n+1 & 3^n-1 \\ 3^n-1 & 3^n+1 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \\ &= \frac{1}{2}\begin{pmatrix} 2(3^n+1)+(3^n-1) & (3^n+1)+2(3^n-1) \\ 2(3^n-1)+(3^n+1) & (3^n-1)+2(3^n+1) \end{pmatrix} \\ &= \frac{1}{2}\begin{pmatrix} 3\cdot 3^n+1 & 3\cdot 3^n-1 \\ 3\cdot 3^n-1 & 3\cdot 3^n+1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 3^{n+1}+1 & 3^{n+1}-1 \\ 3^{n+1}-1 & 3^{n+1}+1 \end{pmatrix}, \end{aligned}$$

which is $P(n+1)$. By the principle of mathematical induction, the formula holds for all $n \geq 1$. [5 marks: checks 2, induction step 2, conclusion 1]

4. Applying $T$ ten times to $(1,0)$ gives $A^{10}\binom{1}{0}$, the first column of $A^{10}$:

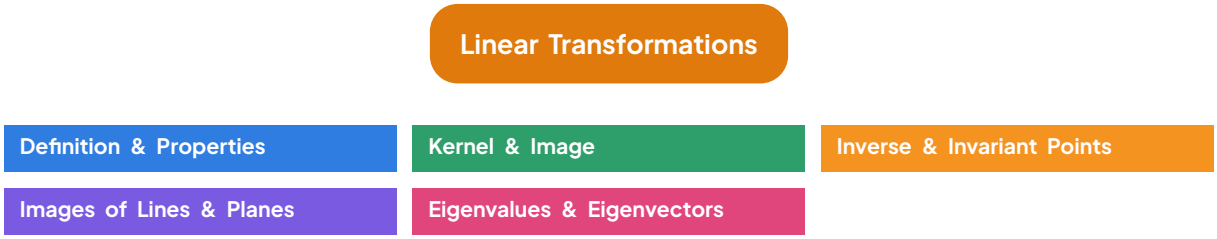
$$A^{10}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 3^{10}+1 \\ 3^{10}-1 \end{pmatrix}.$$

Since $3^{10} = 59049$:

$$T^{10}(1,0) = \left(\frac{59050}{2},\ \frac{59048}{2}\right) = (29525,\ 29524).$$

[3 marks: method 1, value of $3^{10}$ 1, answer 1]

# 21 SUMMARY SHEET




Mind map of the chapter: the five big ideas around linear transformations.

**KEY POINTS** **Summary Sheet — Linear Transformations**

Key definitions

- A map $T : \mathbb{R}^n \to \mathbb{R}^m$ is a **linear transformation** if for all vectors $\vec{u}, \vec{v} \in \mathbb{R}^n$ and all scalars $\lambda, \mu \in \mathbb{R}$: $T(\lambda\vec{u} + \mu\vec{v}) = \lambda T(\vec{u}) + \mu T(\vec{v})$. Equivalently, $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$ and $T(\lambda\vec{u}) = \lambda T(\vec{u})$.
- Every linear transformation has a unique **matrix** $A$ (relative to the standard bases) such that $T(\vec{x}) = A\vec{x}$; the $j$-th column of $A$ is $T(\vec{e}_j)$, the image of the $j$-th standard basis vector.
- **Kernel**: $\ker T = \{\vec{x} \in \mathbb{R}^n : T(\vec{x}) = \vec{0}\}$. **Image**: $\operatorname{Im} T = \{T(\vec{x}) : \vec{x} \in \mathbb{R}^n\} = \{A\vec{x} : \vec{x} \in \mathbb{R}^n\}$ (the column space of $A$).
- $T$ is **invertible** (bijective) iff $\ker T = \{\vec{0}\}$ iff $\det A \neq 0$; then $T^{-1}$ exists and has matrix $A^{-1}$.
- **Invariant (fixed) points**: points $\vec{x}$ such that $T(\vec{x}) = \vec{x}$, i.e. $(A - I)\vec{x} = \vec{0}$.
- **Eigenvalue** $\lambda$ and **eigenvector** $\vec{v} \neq \vec{0}$: $A\vec{v} = \lambda\vec{v}$, found from the **characteristic equation** $\det(A - \lambda I) = 0$.

Formulas and theorems to know

- $T(\vec{0}) = \vec{0}$ always; if $T(\vec{0}) \neq \vec{0}$ the map is not linear.
- Composition: if $T : \mathbb{R}^n \to \mathbb{R}^m$ has matrix $A$ and $S : \mathbb{R}^m \to \mathbb{R}^p$ has matrix $B$, then $S \circ T$ has matrix $BA$ (matrix of $S$ times matrix of $T$) — order matters.
- Kernel and image are **subspaces** of $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively; $\ker T = \{\vec{0}\} \iff T$ is one-to-one (injective).
- Rank–nullity theorem: $\dim(\ker T) + \dim(\operatorname{Im} T) = n$ (dimension of the domain).
- For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$: $\det A = ad - bc$, $A^{-1} = \dfrac{1}{\det A}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (provided $\det A \neq 0$).
- Characteristic polynomial ($2 \times 2$): $\lambda^2 - (\operatorname{tr} A)\lambda + \det A = 0$, where $\operatorname{tr} A = a + d$.

- Eigenvectors for $\lambda$: solve $(A - \lambda I)\vec{v} = \vec{0}$; the set of solutions (including $\vec{0}$) forms the **eigenspace** $E_\lambda$.
- Image of a line through $\vec{a}$ with direction $\vec{d}$: line through $T(\vec{a})$ with direction $T(\vec{d})$ (if $T(\vec{d}) \neq \vec{0}$); if $T(\vec{d}) = \vec{0}$ the line collapses to the point $T(\vec{a})$.
- Image of a plane $\vec{r} = \vec{a} + \lambda\vec{u} + \mu\vec{v}$: plane through $T(\vec{a})$ spanned by $T(\vec{u})$, $T(\vec{v})$ (if they are linearly independent); otherwise a line or a point.

Methods

1. Test linearity: check additivity and homogeneity, or the combined condition $T(\lambda\vec{u} + \mu\vec{v}) = \lambda T(\vec{u}) + \mu T(\vec{v})$; one counterexample disproves linearity.
2. Find the matrix: compute $T(\vec{e}_1)$, $T(\vec{e}_2)$, … and write them as columns of $A$.
3. Kernel: solve $A\vec{x} = \vec{0}$ by Gaussian elimination; give a basis. Image: the column space of $A$; reduce columns to a linearly independent set (pivot columns) to obtain a basis.
4. Inverse: verify $\det A \neq 0$, compute $A^{-1}$ (adjugate/determinant or row reduction), then $T^{-1}(\vec{x}) = A^{-1}\vec{x}$.
5. Invariant points: solve $(A-I)\vec{x} = \vec{0}$; interpret geometrically (fixed point, fixed line, whole space fixed if $A = I$).
6. Eigenvalues/eigenvectors: write $\det(A - \lambda I) = 0$, solve for $\lambda$, then for each $\lambda$ solve $(A - \lambda I)\vec{v} = \vec{0}$ to find a basis of the eigenspace.

Common mistakes

- Forgetting to check $T(\vec{0}) = \vec{0}$ — translations are not linear.
- Reversing the order in composition: the matrix of $S \circ T$ is $BA$, not $AB$.
- Claiming invertibility without checking $\det A \neq 0$ (or $\ker T = \{\vec{0}\}$).
- Solving $(A - \lambda I)\vec{v} = \vec{0}$ with $\vec{v} = \vec{0}$: the zero vector is never an eigenvector.
- Sign errors in $\det(A - \lambda I)$; expand carefully and factorise.
- Confusing invariant points ($T(\vec{x}) = \vec{x}$) with invariant lines (eigenspaces with $\lambda \neq 1$).
- Forgetting that the image of a line may collapse to a point when the direction vector lies in the kernel.

GCE tips

- State definitions precisely — examiners award marks for $T(\lambda\vec{u} + \mu\vec{v}) = \lambda T(\vec{u}) + \mu T(\vec{v})$.
- Show the characteristic equation explicitly before solving; quote $\lambda^2 - (\operatorname{tr} A)\lambda + \det A = 0$ for $2 \times 2$ speed.

- Always give eigenvectors in a simple form, e.g. $\begin{pmatrix}1\\2\end{pmatrix}$, and state the eigenspace as a span.
- For images of lines/planes, write parametric forms first, then apply $T$ to each piece.
- Check answers: verify $A\vec{v} = \lambda\vec{v}$ numerically — a quick safeguard worth marks.

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